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Modeling the Divide: Mathematical Approaches to Understanding and Mitigating Political Polarization

Parth Goyal

School of Mathematics and Statistics
University of St Andrews

I certify that this project report has been written by me, is a record of work carried out by me, and is essentially different from work undertaken for any other purpose or assignment

Abstract

This dissertation delves into the dynamics of political polarization through Agent-Based Modeling (ABM), offering a novel perspective on how individual interactions culminate in societal divides. Employing a methodologically rigorous ABM approach, the study illuminates the roles of exposure, elite influence, responsiveness, and economic inequality in driving polarization. Sensitivity and robustness analyses reveal the nuanced effects of these factors, with independents playing a crucial role in modulating polarization dynamics. The research highlights the significance of initial conditions and the potential for skewed opinion distributions to influence polarization outcomes. By identifying limitations and proposing future extensions, this study not only advances our understanding of polarization but also sets the stage for further exploration in diverse contexts and through advanced computational techniques. This dissertation contributes to the broader discourse on political polarization, suggesting pathways for mitigation and offering insights into the complex interplay of factors that drive societal divisions.

Contents

1	Introduction	5
1.1	Motivation & Context	5
1.2	Literature Review	7
1.2.1	Evolutionary Game Theory (<i>EGT</i>)	8
1.2.2	Graph Theory and Networks	9
1.2.3	Agent Based Modelling (<i>ABM</i>)	10
1.2.4	Discussion of Models	11
2	Methodology	13
2.1	What makes a good ABM?	13
2.2	Agent Assumptions and Methods	13
2.3	Interaction Assumptions and Methods	18
3	Results	21
3.1	Sensitivity Analysis	22
3.2	Robustness Analysis	27
3.3	Dispersion & Model Analysis	29
4	Discussion & Conclusion	30
4.1	Key Takeaways	30
4.2	Limitations & Extensions	31
4.3	Conclusion	33
A	Sensitivity Analysis	34
B	Robustness Analysis #1	42
C	Robustness Analysis #2	48
	Bibliography	52

Chapter One

Introduction

1.1 Motivation & Context

There have been several monumental events over the past 5 years. Coronavirus has speed up transition to digital world and social media. The Russia-Ukraine & Israel-Palestine conflict highlights the deep geopolitical divide between countries and its impact on citizens across the world. Lastly, the economic uncertainty and interconnectedness globally risks a cascading negative affect across various domains.

Consequently these circumstances have accelerated many concerning socio-economic trends, amongst which polarization in society is the worst. [Lukianoff and Haidt, 2019] shows an increase in "safteyism" amongst the younger generation (*Born post 1995 according to them*), which is leading to a lower tolerance to opposing ideological views, especially in college campuses in the US. [Urban, 2023] also brings attention to similar concerns, and how this extends to broader society, highlighting how when tribalism kicks in it can deeply divide people into groups without a rational reasons.

Prominent electoral studies research institutes also confirm these findings. According to [ANES, a] even amongst polarization, party or political polarization tends to be the worst as shown in figure 1.1. Figure 1.2 depicts an increasing negative feelings towards the other party.

Variation in opinion is usually important for a democratic society, as the exchange of ideas and debate is important for truth seeking, however political polarization is not simply a variation of opinion. We define political polarization in two categories [Mccarty, 2019]:

- Ideological: *distribution of political opinions becomes bimodal and deviation between them is high*
- Affective: *Intensified emotional division between members of political parties driven by partisan identities*

Average gap in the share taking a conservative position across 10 political values, by key demographics

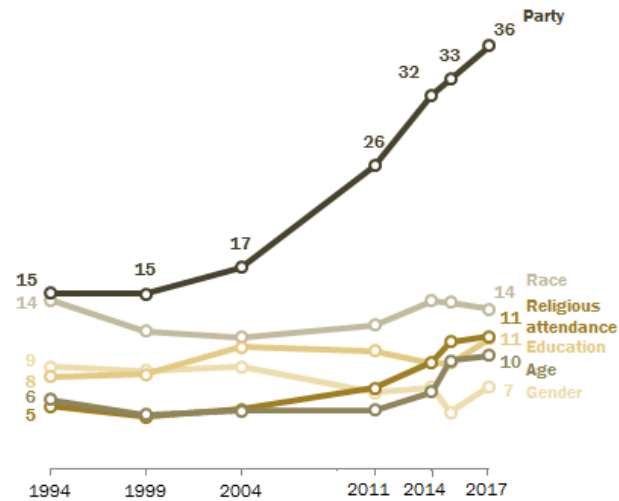


Figure 1.1: Polarization over different topics [Center, 2017]

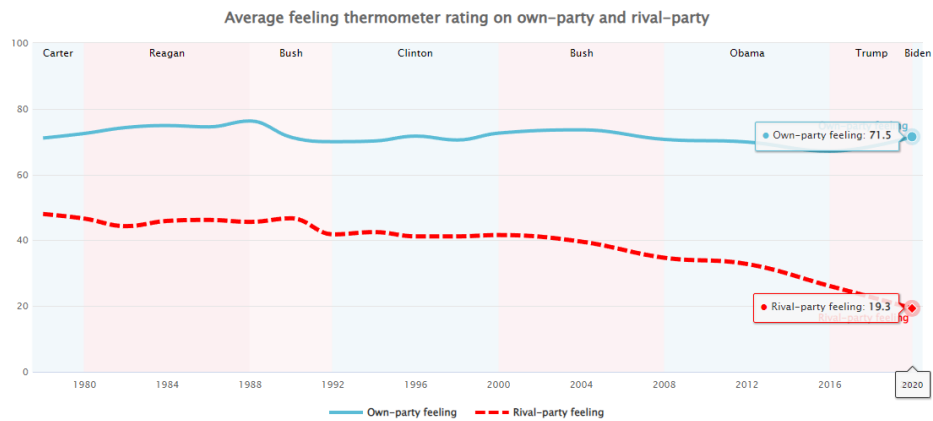


Figure 1.2: Affective Polarization [ANES, a]

These are extremely concerning because political polarization undermines democratic processes by fostering an environment where compromise becomes increasingly difficult. It leads to a decline in the quality of governance, as policymakers prioritize party loyalty over the public good. This environment exacerbates societal divisions, eroding social cohesion and undermining trust in democratic institutions [Skytte, 2020]. It also stifles constructive public discourse, limiting the exchange of ideas and inhibiting policy innovation. Consequently, it hampers the ability of democratic systems to adapt to new challenges, ultimately threatening the stability and functionality of societies [Klein, 2020]. This is especially concerning given the fact that in 2024 almost half the global population is heading into national election [Conte, 2024].

This presents the need for addressing this problem and understand it's dynamics along with potential solutions. There are several accounts of this in the field of psychology, ethnography and their social sciences, however they tend to infer causality in a qualitative manner, and aren't dynamic in nature. Moreover, doing large census collection like the ANES or Pew research center is extremely time consuming and is a lagging indicator.

1.2 Literature Review

The existing literature in the field is quite diverse in its approaches to this problem. There are several interdisciplinary motivated models, that aim to understand the interaction dynamics of political opinion, and it's collective implication. We categorise them into 3 broad models:

- Evolutionary Game Theory - [Stewart et al., 2020][Nowak, 2006]
- Graph Theory and Network Science - [DeGroot, 1974][Conjeaud et al., 2024]
- Agent Based Modelling - [Axelrod et al., 2021] [Kaufman et al., 2024]

This is not an exhaustive categorisation but covers the main approaches to modelling this problem, and are also the approaches that served as our motivation for our choice. We will give an overview of the methodology of one paper from each of these categories, this will highlight different perspective on the field, help understand pros and cons of each type of modelling. It also helps us elaborate and inform our rationale for choosing agent based modelling

1.2.1 Evolutionary Game Theory (EGT)

The model developed by [Stewart et al., 2020] explores the dynamics of political polarization, assuming individuals engage in interactions that are shaped by both in-group and outgroup affiliations, impacted by socio-economic factors. Here probability $\mathbf{p}$ defines an individual's preference for in-group over out-group interaction, where represents $\mathbf{p} = \mathbf{1}$ maximum polarization and $\mathbf{p} = \mathbf{0}$ represents minimum polarization. It assumes an evenly split society which is reasonable considering a US context.

An individual $\mathbf{i}$'s utility function is S-shaped, mathematically denoted as:

$$U(n, l_i, b_i, l_o, b_o, \theta) = S(n, l_i, b_i, l_o, b_o, \theta; \alpha, h) \quad (1.1)$$

$\mathbf{n}$ is the number of interactions, $\mathbf{l}_i, \mathbf{l}_o$ are the number of successful in-group and out-group interactions, $\mathbf{b}_i, \mathbf{b}_o$ represent the benefits derived from these interactions, and θ signifies economic inequality. Parameters α, h adjust curvature and sensitivity allowing to look at shifts in risk aversion as the economic environment fluctuates

The model assumes that out-group interactions are less likely than in-group interactions, but yield higher benefits. This assumption underscores a positive influence model where economic inequality (*measured by gini coefficient*), influence individuals' propensity to engage in risk-averse behaviors.

The overall strategy probability is defined by a multinomial distribution:

$$\pi(k_i, l_o | n) = \binom{n}{k_i} p^{k_i} (1-p)^{n-k_i} \binom{n}{l_o} q_i^{l_o} (1-q_i)^{n-l_o}, \quad (1.2)$$

And the expected benefit per individual is defined as:

$$\sum_{k_i=0}^n \sum_{l_o=0}^n \pi(k_i, l_o | n) (B_i l_i + B_o l_o) = n B_i q_i p + n B_o q_o (1-p) \quad (1.3)$$

In equation (1.2) $\mathbf{p}$ is the strategy and probability an individual chooses a in-group interaction, $\mathbf{q}_i, \mathbf{q}_o$ are the probabilities of successful in-group or out-group interactions, $\mathbf{k}$ is the number of attempted in-group interactions, $\mathbf{n} - \mathbf{k}$ is the total number of attempted out-group interactions. This is translated to get a expected benefit function in equation (1.3).

Having defined the strategies and utility, the dynamic evolution is defined as a copying process, where an $\mathbf{f}$ individual is chosen at random, from a population size $\mathbf{N}$, a second individual is $\mathbf{g}$ is chosen at random to observe. The probability of copying is inversely related to the difference in the expected utility (*fitness*) $\mathbf{w}_g - \mathbf{w}_f$. Expected utility is defined as the product of equation (1.2) and equation (1.1).

Each agent is assumed to optimise for their expected utility by altering their strategy to maximise their expected utility, hence the solution yields is to $\mathbf{p}^*$. This is too complicated to solve dynamically, hence is derived numerically.

The utility functions in EGT models make it much easier to model what is important to an individual, allowing us see more clearly how socio-economic factors inequality can affect the process. However, it makes it very hard to have heterogeneity of agents and the EGT framework demands a strong rationality assumption of maximising a complicated utility function albeit relaxed by the copying process. The analytical appeal breaks down as we add complexity to the model requiring numerical approximations.

1.2.2 Graph Theory and Networks

[DeGroot, 1974] leveraging graph theory and network dynamics formalizes the process through which individuals within a network adjust their opinions through iterative interactions, aiming for a collective agreement or consensus.

Consider a group of k individuals, each holding a subjective probability $F_i(\theta)$ distribution over some parameter θ , reflecting their initial opinion or belief about a certain issue. The objective is to model the evolution of these individual beliefs through mutual interactions within the group.

The update of opinions is modelled through a weighted linear combination of the current opinions in the group. For the n -th iteration of opinion updates the individual i 's opinion is updated as:

$$F_i^{(n+1)} = \sum_{j=1}^k p_{ij} F_j^{(n+1)} \quad (1.4)$$

where p_{ij} represents the weight that individual i assigns to individual j 's opinion during the n -th iteration, reflecting the underlying network structure and dynamics of influence among individuals

The weights p_{ij} reflect is essentially a $k \times k$ transition matrix in a Markov Process. The sum of all the elements in a row equal 1. An example of transition matrix $\mathbf{P}$ for a 2 agent model is:

$$\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{3}{4} & \frac{1}{4} \end{bmatrix}$$

In this matrix agent 2 places a higher weight on agent 1's opinion, whereas for agent 1 equally weights opinions between them. This can be extended for several number of agents and represents the updating mechanism.

The iterative process of updating opinions continues until the opinions converge mathematically represented by finding a stationary distribution $\mathbf{F}^*$ such that:

$$\mathbf{F}^* = \mathbf{P}\mathbf{F}^* \quad (1.5)$$

The transition matrix $\mathbf{P}$ serves as a critical mathematical abstraction of the influence network amongst individuals, dictating the dynamics of opinion formation and evolution. In [DeGroot, 1974] only consensus formation is considered, however there are several extensions such as [Conjeaud et al., 2024], which include stubbornness and changes to opinion based on public opinion.

Graph based model, allow us to understand the mathematical properties of convergence and divergence easily by looking at the transition matrix, and is more flexible in nature than EGTs, however it makes it hard to include stochasticity in the model, assumes a formal structure for social interaction and is mathematically complicated when trying to include heterogeneity in agents. Moreover, over large number of individuals the computation complexity increases drastically.

1.2.3 Agent Based Modelling (*ABM*)

Agent-Based modelling offers a nuanced approach to understand social phenomenon. [Axelrod et al., 2021] is a prime example, providing insight into how individual interactions based on ideological positions can lead to societal polarization over time.

In the model, each agent i 's ideological position is represented by a point in a one-dimensional space $\mathbf{x}_i$, which is generated from a normal distribution. The dynamics of the model are driven by two fundamental processes: interaction probability and opinion updating mechanism.

The interaction probability is defined as:

$$p = \frac{1}{2} \frac{d}{E} \quad (1.6)$$

where d is simply the absolute difference in opinion between 2 agents and E (*Exposure*) is a sensitivity parameter. The intuition is that the further you are ideologically from an individual the less likely you are to interact with them

The opinion update mechanism is a mix of positive and negative influence [Smaldino, 2023]. It displays the importance of homophily in social science, which is the tendency for similar individuals to preferentially assort and interact [Smaldino, 2023]. Whenever d between two agents is within a tolerance level T , it is an attraction (*positive influence*)

and if it's beyond T then its repulsion (*negative influence*). The opinions update in the following way for agent i at time t :

$$x_i(t+1) = \begin{cases} x_i(t) + R * (x_j(t) - x_i(t)) & \text{if attraction} \\ x_i(t) - R * (x_j(t) - x_i(t)) & \text{if repulsion} \end{cases} \quad (1.7)$$

R (*Responsiveness*) here is another sensitivity parameter moderating the impact on an agent's opinion.

Based on these rules of the environment, several simulations are run to assess emergent properties. In figure 1.3 you can see how over several steps (*attempted interactions*) the distribution of opinions changes. In this method polarization is measured as variance of the opinions and the modality. It can clearly be seen how over several simulations at $T = 0.25$, the deviation in opinion is increasing, polarization increases, whereas when we increase Tolerance level there is a convergence in opinions.

ABMs don't have distinct mathematical properties that can be examined, but offer a lot of flexibility, are computationally easier, have an element of stochasticity that is appealing for modelling social behaviour and is emergent, hence makes tracking the changes a lot easier over time.

1.2.4 Discussion of Models

There are several many ways to model social behaviour such as the Schelling segregation or complex contagion models to name a few. We have highlighted these particular models and papers as they broadly reflect a class of models most commonly used, and to briefly elaborate how they are implemented.

Post considering these class of models We decided to examine the problem of political polarization using ABMs. For the purpose of creating your own model, they offer the most flexibility, and it's easier to add sophistication in interactions at low analytical complexity. It's easier to link assumptions to research in social sciences i.e. "input realism" and include heterogeneity amongst agents [Manzo, 2022]. There is the trade off that it's not backed by strict mathematical proofs and analytical solutions like the other methods, but as a modeller's choice We believe the trade off is appropriate for this context.

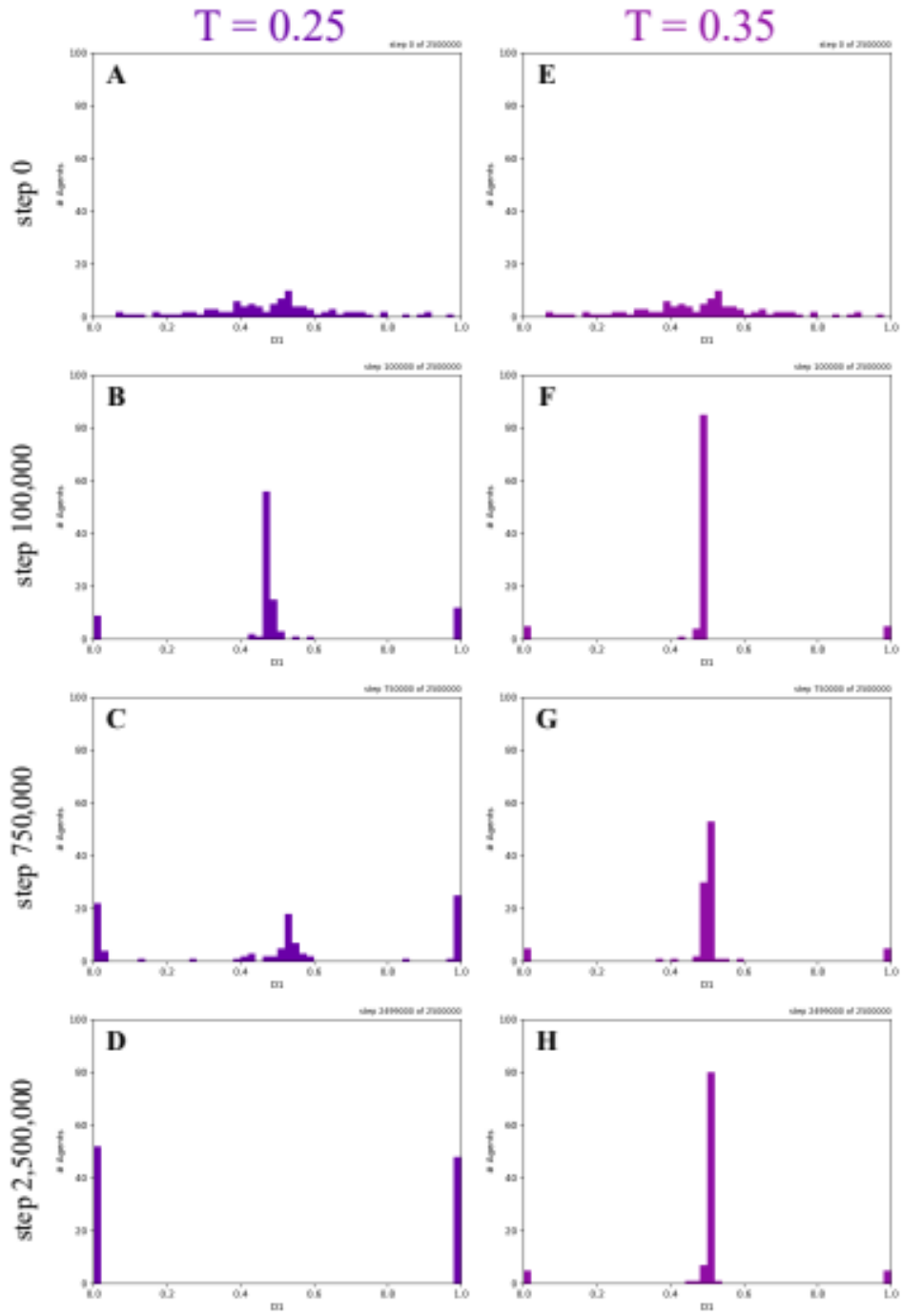


Figure 1.3: Polarization over steps for different T [Axelrod et al., 2021]

Chapter Two

Methodology

2.1 What makes a good ABM?

ABMs in itself are a broad class of models, and their usefulness depends deeply on the rules that govern them and how agents are defined. [Manzo, 2022] goes into detail about what makes an ABM useful for causal inference in a social context, and it proposes the following criteria:

- Theoretical Realism - *Assumptions should be backed by sociological/psychological theories and evidence*
- Input Realism - *Initialisation of the model parameters and functions should be based on real data*
- Output Realism - *Validation of the model against specific datasets for realistic outcomes*

A good ABM should at minimum have theoretical and input realism to reliably explain to explain social behaviour, with some leniency on output realism as social systems tend to have a mapping problem defined as the potential mismatch between components of one's models on one hand and the measurements that one makes of empirical systems [Smaldino, 2023]. Unlike subjects like physics, in social context it's not clear what are the measurements and their 'units', the aim is to get to a proxy that accurately reflects the qualitative situation in the real world. Keeping this in mind we will elaborate on the key assumptions of our model and how it falls under the realism framework.

2.2 Agent Assumptions and Methods

An agent in our model is anyone who has a political opinion, i.e. the literate population. Similar to [Axelrod et al., 2021] we define our parameter of interest or symbolic y variable

for an agent as a political opinion on a continuous one dimension scale between 0 and 1. Many models such as [Braha and de Aguiar, 2017] assume a discrete scale of 0 or 1, but we feel it’s a limiting assumption as its unfair to assume a person’s political opinion is binary. The reason for only considering 1 dimensional parameter space is a modeller choice. It’s a matter of how rational the process of polarization is, anecdotal experience from [Urban, 2023] and shortened interaction times due to social media provide some evidence for limited rationality, hence a stronger case for that an agent weighing only one aggregate opinion scale is an adequate abstraction. This scale is to capture the ideological component of political polarization in our model.

Agents are heterogeneous in our model and have defining socio-economic characteristics other than their parameter of interest. The 2 characteristics are (*these are Boolean*):

- Elite Status - elites have a huge impact on people’s opinion and mass outreach. In a graph based model, this would be represented by a high centrality measure
- Partisan Status - partisans strongly support a party and have potential bias against the opposing party

Elite status is included as a characteristic because there are several papers that debate on the impact of elites on population polarization such as [Druckman et al., 2013]. In our simulation elites will be a stringent minority by the nature of their definition.

Partisan status is included to capture the affective component of political polarization, and is motivated by the idea that individuals in a population already start from a bimodal distribution, a limitation in [Axelrod et al., 2021]. Evidence of bimodality has been highlighted by [Enders, 2021] as seen in figure 2.1. It depicts how masses (*Non-Elites*) and elites distribution on issues (ideological) and thermometer (affective) based ideal points is bimodal in the US. Another interesting observation here is that the elites are more polarized than the masses on both scales.

These assumptions and observations inform the way we initiate our agents. We initiate our agents from the following normal Distributions:

$$x_i(0) \sim \begin{cases} \mathcal{N}(0.5, 0.2^2) & \text{if } i = \text{Independent} \\ \mathcal{N}(0.35, 0.15^2) & \text{if } i = \text{Mass Democrat} \\ \mathcal{N}(0.65, 0.15^2) & \text{if } i = \text{Mass Republican} \\ \mathcal{N}(0.35, 0.2^2) & \text{if } i = \text{Elite Democrat} \\ \mathcal{N}(0.65, 0.2^2) & \text{if } i = \text{Elite Republican} \end{cases} \quad (2.1)$$

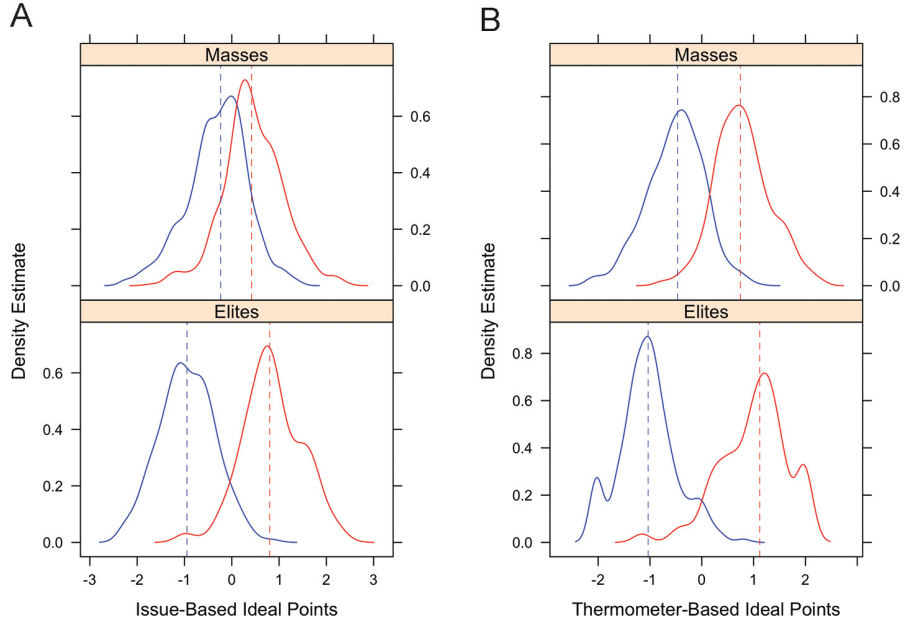


Figure 2.1: Mass vs Elite Polarization [Enders, 2021]

In equation 2.1 the shifted means for republicans and democrats reflect an already affective population, and these are also agents who would have partisan status. To keep the model intuitive left shifted mean reflects democrats and right shifted mean reflects republicans. Within partisan agents, elites have a higher variance parameter to capture fatter tails as shown in figure 2.1, which translate to higher polarization within elite as displayed in [Enders, 2021]. We have not shifted the means more extremely for elites to limit the modality of our initial distribution. Independents are non-partisan class of agents that don't identify strongly with a Party, it's common across democratic nations to have this class of agents, and is well documented in the US context [ANES, a]. We are making a modeller's choice of assuming that independent elites and independent mass have same variance, as We have not come across any empirical or anecdotal evidence to assume otherwise. Figure 2.2 shows what the initial distribution looks like.

Within our model at initiation we need to decide the number or proportion of agents that belong in each category. For partisan status we refer to [ANES, a] in informing our proportions. Figure 2.3 displays how people identified with political parties across the US over time. We make the assumption that independent republicans and independent democrats, can't be assumed to be partisan under the definition we give, this is also confirmed by figure 2.4. The proportions are reflected in table 2.1 and remain fixed in all simulations.

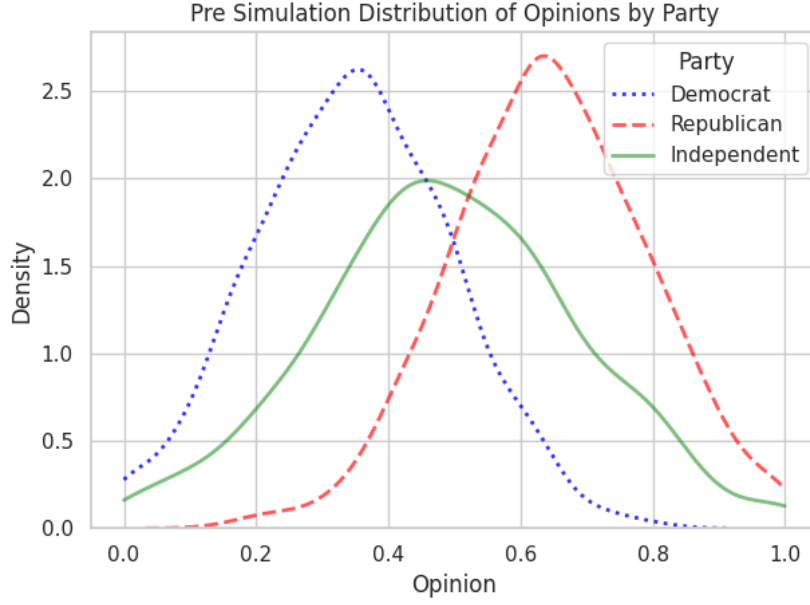


Figure 2.2: Initiation distribution of opinions by party

Party	Proportion
Democrats	33.1%
Independents	33.8%
Republicans	33.1%

Table 2.1: Party Proportions in Simulations

For elites We could not find any good empirical estimate, probably because it is something that can be defined in a number of ways, hence our proportion for this is arbitrary while ensuring its a strict minority. The changing proportion of elites is something that is explored in sensitivity analysis. The specific proportion of elites that fall under various parties however is still scaled by table 2.1, for example if $Elite = 0.5\%$, then $Independent\ Elite = 0.5\% \times 33.8\%$

As seen We have implemented input and theoretical realism, wherever feasible while trying to define our agents. Having described them, We will now move on to explain the assumptions and methods of the interactional component.

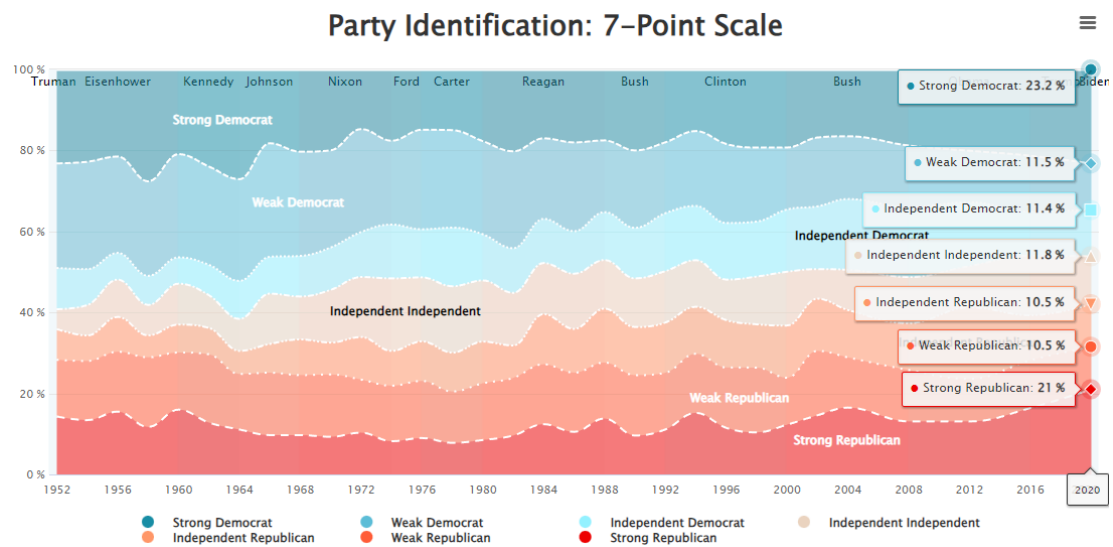


Figure 2.3: Party Identification in the US [ANES, a]

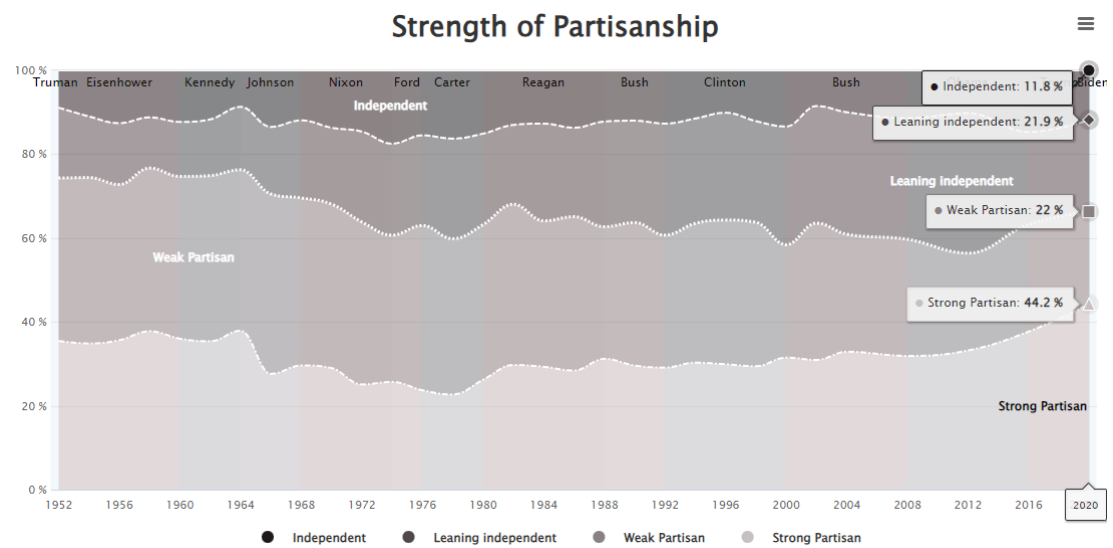


Figure 2.4: Partisan in the US over time [ANES, a]

2.3 Interaction Assumptions and Methods

In our ABM interactions are defined pairwise, as it is easier to code and simulate, also is the most common type of interaction in most ABM literature. Interactions are dependant on the characteristics of agents and are a combination of positive and negative influence. This is motivated by Social Identity Theory and homophily [Turner and Tajfel, 1979]. It highlights bias for "in-group" and favourable treatment towards those are similar to us, while also showcasing favourable comparison of in-group against out-group. Studies such as [Banda and Cluverius, 2018] also showcase evidence strong negative influence especially by elites.

The rule of attraction and repulsion is still governed by tolerance level as described [Axelrod et al., 2021], however our model's process is a sophistication on top of the basic model introduced by [Axelrod et al., 2021]. We will describe some of the assumptions underlying our interaction process.

If partisan elites interact with mass they will have a strong (positive or negative) impact on the mass than when only masses interact, and they don't change their positions when interacting with masses. There are many studies that have seen this trend such as as [Bäck et al., 2023],[Skytte, 2020]. Interactions between same party partisans will have a convergence to the middle and is always of positive influence regardless of tolerance due to tribalism. As a modeller choice interactions involving independents tend to be less polarizing in general as they don't have partisan identity hence there isn't a similar exception as same party partisans. Independent elites tend to have a nuanced interaction, where in attraction scenarios they have strong positive influence, however in repulsion scenarios the influence isn't that strong. This ties back to assumption that independents are less polarizing.

Post initiation as described in table 2.1, each agent goes through all possible pairs of interaction, where then the probability of an interaction is defined by the following formula:

$$p = \begin{cases} \frac{1}{2}^{\frac{d}{E}} & \text{if neither is Elite} \\ \frac{1}{2}^{\frac{d}{E \times multiplier}} & \text{if one or both are Elite} \end{cases} \quad (2.2)$$

Here the parameters d & E have the same meaning as they did in equation 1.6. The probability function is exponential in nature and is inversely related to the distance between two agents. It is reasonable to assume that likelihood of interaction decreases exponentially as differences in opinion increase, this is the very nature of how social media algorithms function and socio-economic segregation in society. The likelihood

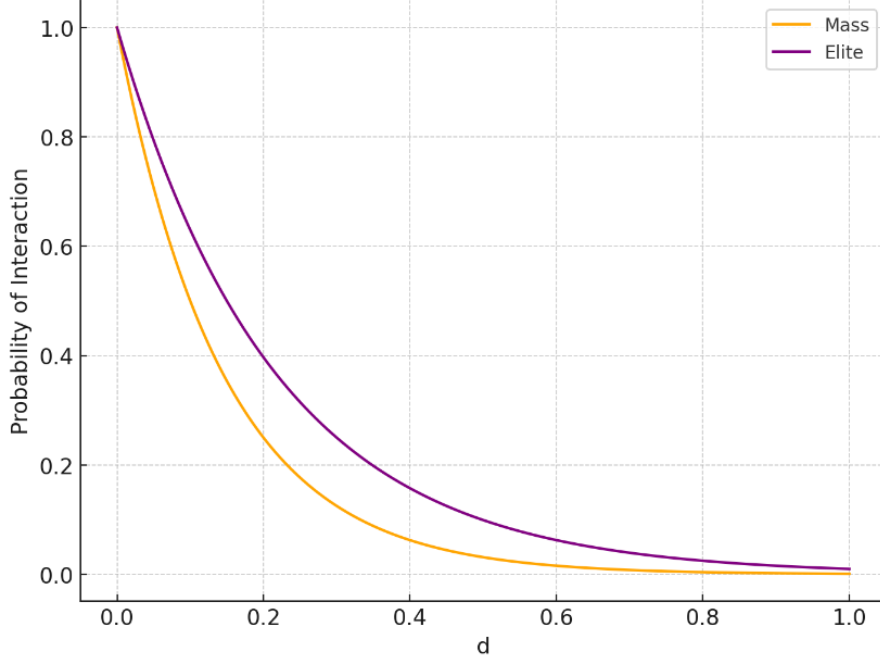


Figure 2.5: Mass vs Elite Probability of Interaction

of interaction is also assumed to be higher when there is an elite on either side of the interaction, this is done by the *multiplier* parameter. This assumption is made because we define elites as people who have broader and denser networks, hence accordingly they will have more interactions in the defined environment. Figure 2.5 depicts the shape of the probability function.

There are essentially 6 types of opinion updates in this model excluding the case where elites don't change their opinion when interacting with masses:

$$x_i(t+1) = \begin{cases} x_i(t) + R \times (x_j(t) - x_i(t)) \times multiplier & \text{strong positive influence} \\ x_i(t) + R \times (x_j(t) - x_i(t)) & \text{regular positive influence} \\ x_i(t) + \frac{R \times (x_j(t) - x_i(t))}{multiplier} & \text{weak positive influence} \\ x_i(t) - R \times (x_j(t) - x_i(t)) \times multiplier & \text{strong negative influence} \\ x_i(t) - R \times (x_j(t) - x_i(t)) & \text{regular negative influence} \\ x_i(t) - \frac{R \times (x_j(t) - x_i(t))}{multiplier} & \text{weak negative influence} \end{cases} \quad (2.3)$$

Equation 2.3 summarises all the possible opinion updates possible in our model. The potential conditions for each type of influence can be briefly summarised by table 2.2.

Influence Type	Agents and Tolerance Level
Strong positive influence	• Same party partisans, an Elite & Mass
Regular positive influence	• Same party partisans, same elite status • Both Independents, same elite status within tolerance • Different party partisans, an Elite & mass within tolerance • Partisan Elite & Independent Mass, within tolerance
Weak positive influence	• Same elite status, different party partisans, within tolerance
Strong negative influence	• Different party partisans, an Elite & Mass, outside tolerance
Regular negative influence	• Different party partisans, same elite status, outside tolerance • Partisan Elites & Mass Independent, outside tolerance
Weak negative influence	• Both Independents outside tolerance

Table 2.2: Conditions for different Influence Types

Tolerance is not being arbitrarily defined in our model and has some economic significance. Our tolerance measure is scaled with the Gini coefficient; a measure of economic inequality in society, where 0 represents perfect equality and 1 represents perfect inequality. The reason being that inequality and perception of it affects people’s attitude towards politics in a directional manner i.e. higher inequality is accompanied by poorer feelings towards politics [equalitytrust.org.uk].

$$Tolerance = \frac{T}{GiniCoefficient} \quad (2.4)$$

In equation 2.4 T is a sensitivity parameter to ensure the tolerance level is reasonable. Gini is inversely related to tolerance levels in society. We will be using $Gini = 0.434$ as base case in our model as that’s the latest estimate for the US [ANES, b], but will vary it in the next section.

We have implemented input and theoretical realism, wherever feasible while defining agent interactions, also keeping the overall model relatively intuitive. Having described the rules of our ABM, which create our environment, several simulations are then run to understand the emergent properties of our model. These are discussed in the next chapter.

Chapter Three

Results

In this chapter we explore the findings of our ABM model over the course of several simulations. Firstly, its important to test our model for the purpose of making useful inference, once we have established the level of uncertainty in our model then we can go on to make an assessment of the key takeaways in our discussion. As mentioned in the previous chapter for the purpose of "output realism" the model needs to be validated against specific datasets, however due to the social mapping problem it's hard to have clear outputs to measure against in context of political polarization, hence as a potential alternative [Manzo, 2022] suggests the following analysis of the underlying model:

- Sensitivity Analysis: *assessment of variability of simulated outcomes due the changes in the model's parameter values*
- Robustness Analysis: *assessment of variability of simulated outcomes due to the changes in the model's internal details*
- Dispersion Analysis: *assessment of variability of simulated outcomes across repeated simulation under same parameter values*
- Model Analysis: *analysis of how the ABM works internally once the simulation is launched*

We will address the sensitivity and robustness analysis in detail and with a reasonable argument show why we can assume that our model is relatively to immune from discrepancy in dispersion and model analysis. We want to highlight due to the nature of the analysis, it's almost impossible to have an exhaustive analysis across all parameter spaces, hence as a modeller the analysis has been limited at discretion to ensure the main takeaways are clearly understood. Accordingly, any limitations due to this will be addressed in the discussion section. Before jumping into the simulations We want to establish a base case, from which our simulation is initiated until stated otherwise.

There are fixed parameters and functions in our model which won't change throughout due to it's tie to input realism and ensuring complexity isn't unnecessarily high. Proportion within each party will remain constant throughout the simulations, the opinion update core function will remain the same and our sensitivity analysis will be conducted assuming *centrus paribus*.

The initial distribution in equation 2.1, is our base for agent sampling. The parameters have the following the base case:

Parameter	Base Value
N	5000
E	0.1
T	0.1
R	0.25
Gini Coefficient	0.434
Impact multiplier	1.2
Prob multiplier	1.2

Table 3.1: Base case values for parameters

Our number of agents is much orders of magnitude larger than most other literature, this to reduce the dispersion in simulations, while maintaining low runtime for sensitivity and robustness analysis. E , T , R and the multipliers are arbitrary, but are set in a way where they are approximately the average value in our sensitivity analysis parameter space. Gini Coefficient is set as the estimate from [ANES, b].

3.1 Sensitivity Analysis

Figure 2.2 shows how the distribution of opinion looks like when segregated by party, in aggregation the initiated distribution of opinion is shown in figure 3.1

As we increase the number of simulations, the number of interactions per agents increase, and the bimodality in the graph grows stronger, until it reaches a polarized state at 2000 simulations. This is clearly shown in appendix A, and highlights the relationship that in our model overtime, as agent number of interactions increases the polarization in the population also increase.

Similarly we explore the relationship of various parameters over simulations to understand their impact in this process. We have limited our analysis to 500 simulations to deal with increased run times.

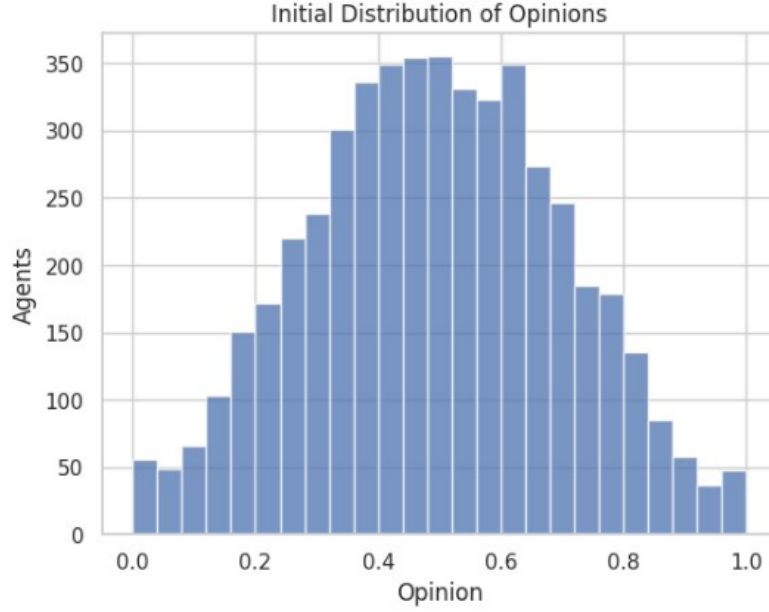


Figure 3.1: Initial distribution of opinions aggregated

Figure 3.2 depicts the rate of change of polarization (*standard deviation of opinions*) for different exposure values over several simulations. Very evidently exposure is directly proportional polarization, especially for exposure values greater than **10.5**, where it reaches a stable state of **0.5** very quickly. This tracks with our previous finding that as number of simulations and interactions increases, polarization increases. Equation 2.2 shows probability is directly proportional to exposure, hence it is reasonable to expect similar properties.

Unlike what you would expect, the responsiveness parameter does not seem to exhibit a clear positive relation with polarization. There are mixed results as R is increased, this trend persists even as the number of simulations increase. It might highlight that some of positive influence behaviour caused by interaction with independents slows the rate of polarization. We can look into if impact multiplier and proportion of elites in the population show similar properties.

Impact multiplier is a parameter that magnifies the opinion update to have strong influence as seen in equation 2.3. Interestingly enough it also displays mixed results like R , where there is not a clear increase in polarization with an increase in the impact of elites, as shown in figure 3.4. This is also confirmed when we look at the end state polarization post 500 simulations for various proportions of elite in figure 3.5, as its also not increasing with an increase in elites. This hints that introduction of independents as

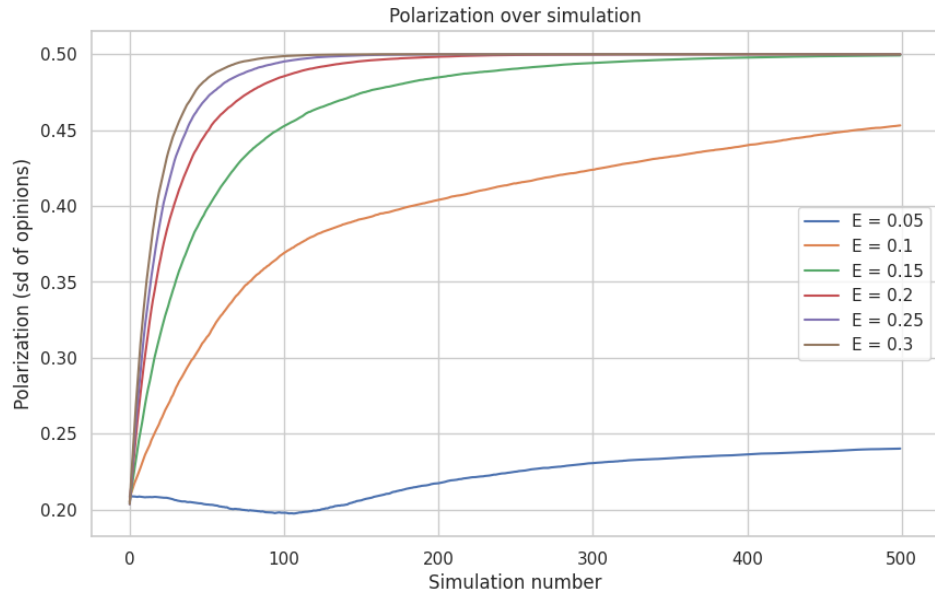


Figure 3.2: Polarization over different Exposure Values

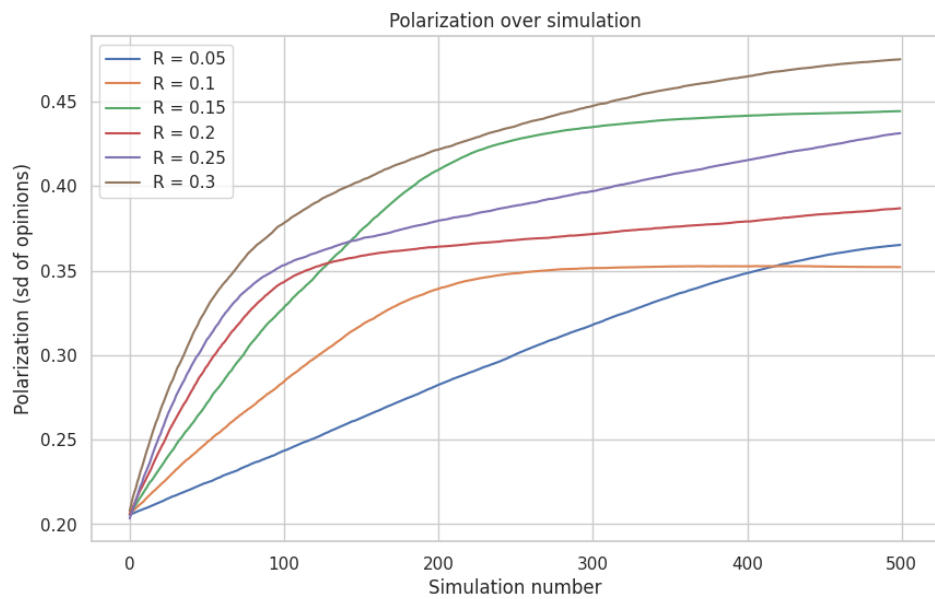


Figure 3.3: Polarization over different Responsiveness Values

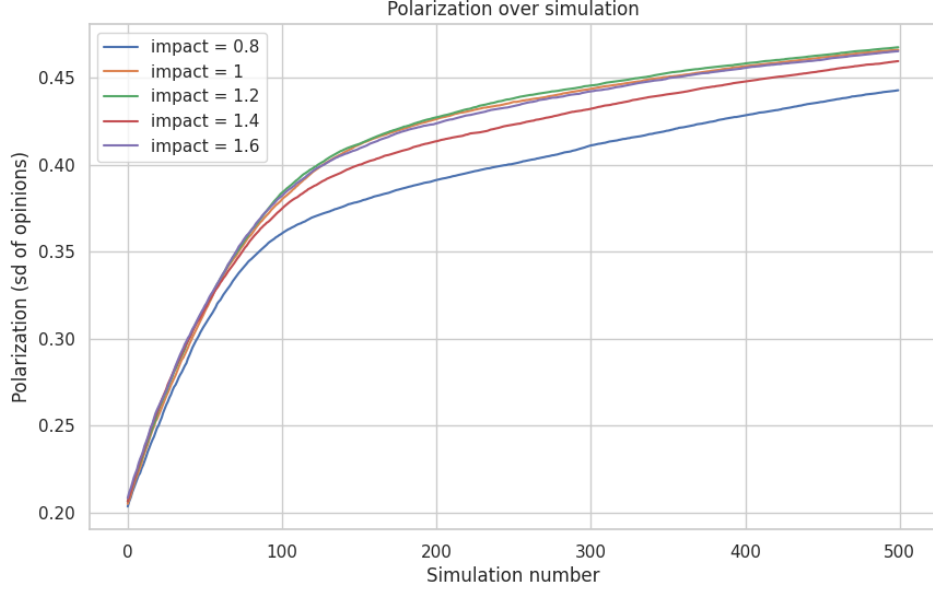


Figure 3.4: Polarization over different Elite Impact Multipliers

agents has a slowing effect on polarization.

Probability multiplier increases the likelihood of interaction for elite agents. It does not have any interesting results as seen in figure 3.6 and seems to track what was observed with the results for the exposure parameter.

Lastly, our most interesting result is for the inequality parameter. We noted that the Gini is inversely related to tolerance in the population in equation 2.4. It translates to having directly proportional impact on polarization, hence lower the inequality leads to lower polarization in society, and potentially depolarizing society if it causes tolerance level to be low enough.

Our sensitivity analysis reveals a nuanced interplay of parameters affecting polarization, with exposure notably driving it. The mixed results from responsiveness and elite influence suggest complexity, softened by independents. The inverse relation between inequality and tolerance highlights inequality's pivotal role in polarization. These insights affirm the model's capacity to reflect real-world dynamics, emphasizing the need to consider multifaceted interactions in tackling societal polarization.

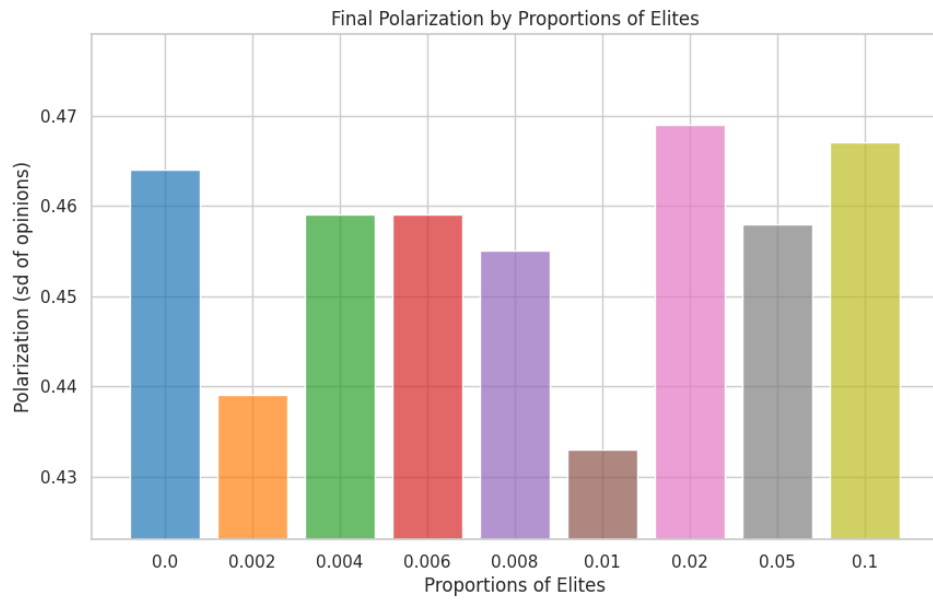


Figure 3.5: Polarization for varying proportion of Elites

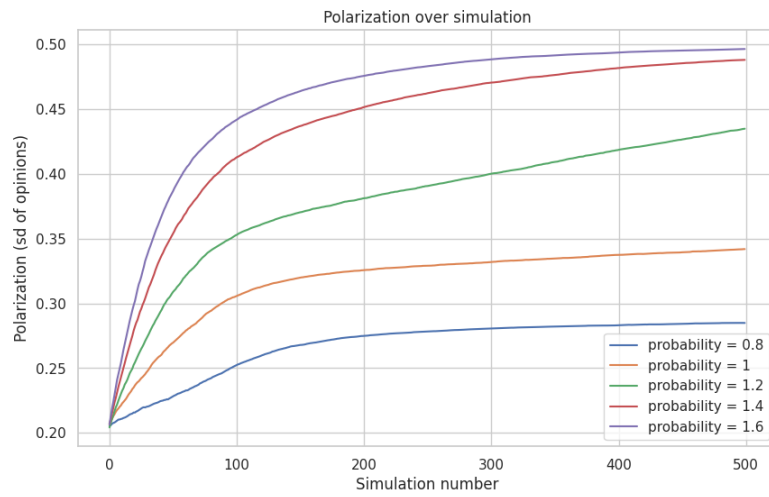


Figure 3.6: Polarization over different Elite Probability Multipliers

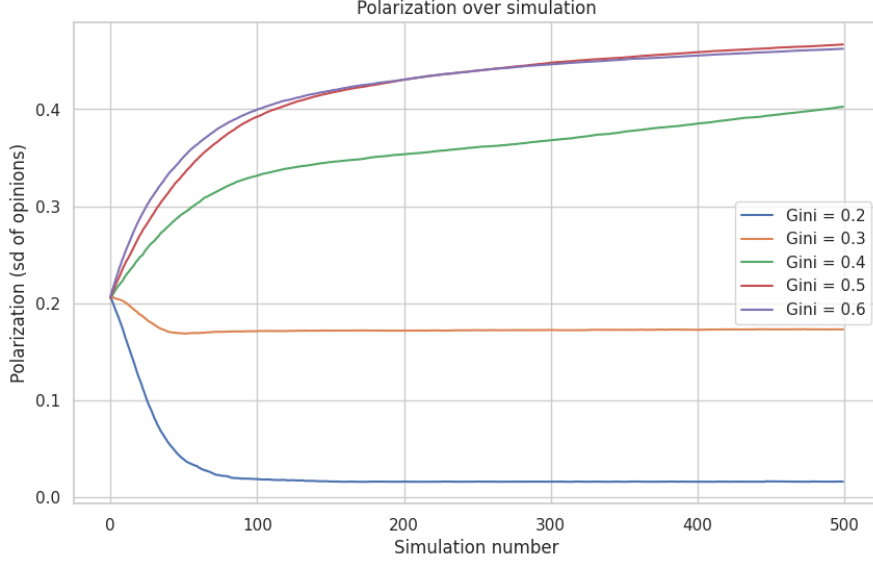


Figure 3.7: Polarization over different Inequality Values

3.2 Robustness Analysis

There are variety of assumptions that can be tested when considering robustness, for our purpose we test our initiation assumptions, this is because we want to observe what occurs when our process starts from different states. We first look into what happens when we reduce the partisanship at the initiation, and tame the polarization of elites. This is achieved by adjusting the initial parameters in equation 2.1 to the following equation 3.1.

$$x_i(0) \sim \begin{cases} \mathcal{N}(0.5, 0.15^2) & \text{if } i = \text{Independent} \\ \mathcal{N}(0.4, 0.15^2) & \text{if } i = \text{Democrat} \\ \mathcal{N}(0.6, 0.15^2) & \text{if } i = \text{Republican} \end{cases} \quad (3.1)$$

Mass and elites are now initialised from the same distribution, this looks like figure 3.8. When initiated like this the rate of polarization dramatically reduces in the population. This can be seen in appendix B. Very evidently even as the number of simulation grows agents move diverge much less, and even at 2000 simulations there still are agents in the center (0.5). Suggesting that from reduced partisanship and affective polarization state it is harder to reach a fully polarized state.

Secondly we aim to explore what occurs if we initialise agents using a different distribution and slightly nudge the population towards a side. This is achieved by

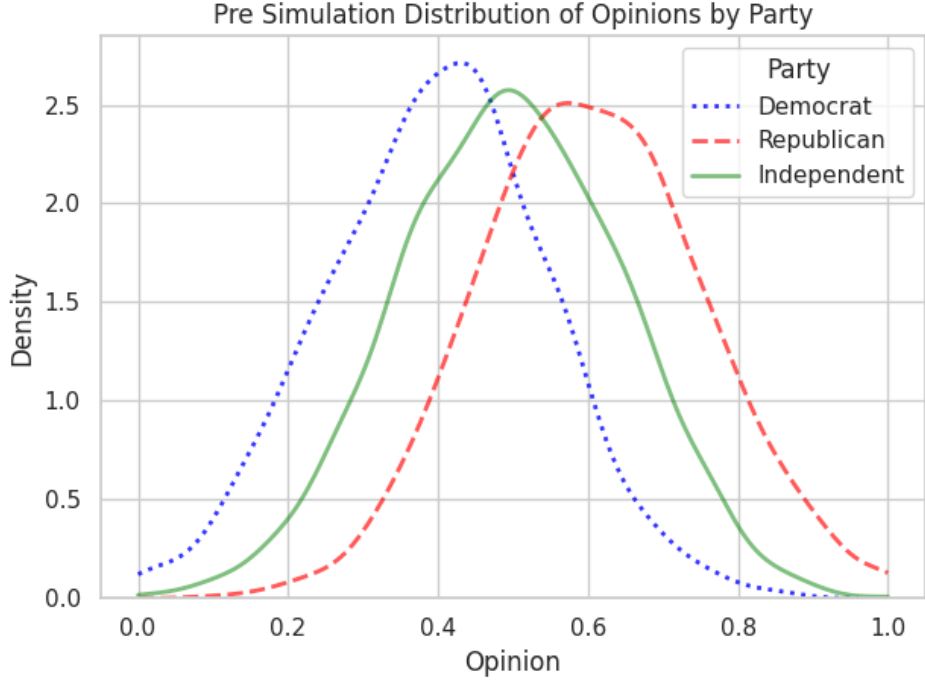


Figure 3.8: Initial distribution of opinions by party (*Less polarized*)

initialising agents using a Beta distribution as depicted in equation 3.2

$$x_i(0) \sim \begin{cases} \mathcal{B}(4.5, 6.0) & \text{if } i = \text{Independent} \\ \mathcal{B}(2.7, 7) & \text{if } i = \text{Mass Democrat} \\ \mathcal{B}(6.8, 4.8) & \text{if } i = \text{Mass Republican} \\ \mathcal{B}(2.5, 6.5) & \text{if } i = \text{Elite Democrat} \\ \mathcal{B}(6.5, 4.5) & \text{if } i = \text{Elite Republican} \end{cases} \quad (3.2)$$

This yields and initial distribution as depicted in figure 3.9. Opposite to the previous situation, when a population is skewed the rate of polarization dramatically increases, this can be seen in appendix C where only after 100 simulations the population is segregated and by 500 simulations it's almost fully polarized. Since we slightly right skew the distributions, the polarized states reflects this as more of the population then ends up getting polarized towards the left.

The robustness analysis provides interesting insights into how initial conditions can have an out sized impact in polarizing a population, it also gives us evidence that our model is robust to varying assumptions while providing useful insights

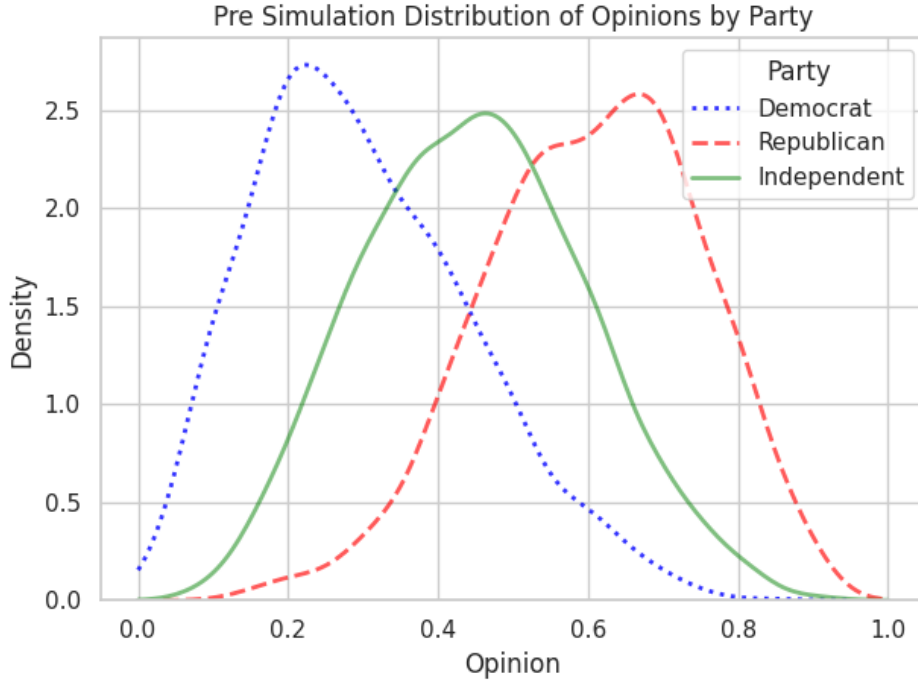


Figure 3.9: Initial distribution of opinions by party (*Skewed*)

3.3 Dispersion & Model Analysis

Over several simulations the trends presented in previous analysis persist due our large number of agents, hence the variability in results is not something we believed were worth addressing. In appendix A the distribution of number of interactions is presented to give a rough idea of how many successful contacts do agents have and it scales roughly linearly with number simulations in our environment, reducing cause to worry about variability. We don't present any strict quantitative relations for our process due to it's social nature, which was another reason dispersion analysis is not a priority, however if need be this can be easily conducted at a later date with our code [ParthG60, 2024].

Model analysis is something that helps depict how the simulation works. Since the methodology is intuitive and straightforward, this was not a major point of research. We don't use any complicated formulas and its quite clear why and how we arrive out our results. Out of curiosity, however there are some Markov methods other researchers employ for this effect and can be considered later.

Chapter Four

Discussion & Conclusion

In this section, we will discuss the key takeaways than can we inferred from our methodology and results, the limitations and potential extensions of this study and concluding. Before you proceed to read to this last bit, we want to leave you with our two favourite quotes that underpin our philosophy to the initial problem described in the introduction and hope as reader you feel this philosophy has been reflected in our work.

- *"All models are wrong, but some are useful"* - George Box
- *"Science is the art of systematic oversimplification"* - Karl Popper

4.1 Key Takeaways

In the exploration of political polarization through the lens of Agent-Based Modeling (ABM), our study has unearthed several pivotal insights that underscore the nuanced interplay of individual behaviors and systemic dynamics. The application of ABMs, grounded in a robust methodological framework that considers theoretical and input realism enabled a comprehensive dissection of the multifaceted phenomena underpinning political polarization. This approach has proven invaluable in simulating complex social interactions, offering granular insights into the aggregation of individual actions into collective outcomes.

Our sensitivity analysis brought to light the critical role of exposure in driving polarization. It highlighted how increased interactions within ideologically cohesive groups serve to amplify existing divides, echoing the real-world dynamics of echo chambers where selective interaction amplifies societal divisions. This finding underscores the imperative to foster cross-cutting dialogues as a countermeasure to polarization. It also ties back to some of suggestions made in [Lukianoff and Haidt, 2019] and [Urban, 2023].

Furthermore, the nuanced outcomes from adjusting parameters such as responsiveness and elite influence reveal the intricate balance of forces shaping societal polarization. The

analysis underscored the significant, yet complex, impact of elite agents, whose influence is both profound and modulated by the presence of independents. These independent agents emerged as vital moderators, tempering the polarizing effects of elite-driven narratives.

A particularly revelatory aspect of our investigation was the analysis of economic inequality’s impact on societal divides. The discovery of an inverse relationship between inequality and societal tolerance not only highlights the profound socio-economic roots of polarization but also signals the necessity of addressing economic disparities as part of any comprehensive strategy to mitigate societal polarization.

The robustness analysis, focusing on the implications of varying levels of partisanship and skewed initial opinion distributions, added another layer of depth to our understanding. Scenarios characterized by lower initial partisanship progressed more slowly towards polarization, suggesting the significance of initial societal conditions in determining the trajectory of polarization. In contrast, scenarios with skewed initial opinion distributions revealed the system’s acute sensitivity to the starting landscape of societal opinions, with significant implications for the speed and extent of polarization.

Collectively, these insights validate the utility of ABMs as a sophisticated tool for modeling the complex dynamics of political polarization. They highlight the necessity of considering a broad spectrum of interactions, initial conditions, and systemic factors. This comprehensive analysis not only reflects the model’s capacity to mimic the intricacies of real-world dynamics but also emphasizes the importance of adopting multifaceted, interdisciplinary approaches in understanding and addressing the challenge of societal polarization.

4.2 Limitations & Extensions

In our exploration of political polarization through Agent-Based Modeling (ABM), we’ve navigated complex social dynamics with a model that simplifies these interactions into a one-dimensional ideological space. While this approach facilitates computational efficiency and conceptual clarity, it may not fully capture the multidimensional nature of political beliefs and identities. This simplification underscores a critical limitation of our model, potentially offering a reductive perspective on political alignments that fails to account for the nuanced spectrum of real-world ideologies.

Moreover, our model’s landscape is devoid of external shocks—unpredictable, significant events that can dramatically sway public opinion and polarization levels. The absence of such dynamics limits the model’s capacity to simulate societal responses to real-world incidents, marking a notable gap in our exploration of political polarization.

Additionally, the breadth of our sensitivity analysis, though extensive, does not encompass global parameters or a broader spectrum of model assumptions. This restriction narrows our investigation into the model’s robustness and its sensitivity to an expanded set of variables, suggesting areas for future refinement.

Another area where our study seeks expansion is in its empirical grounding. While informed by theoretical insights and empirical observations, our model lacks direct validation against real-world data, raising questions about its applicability and precision in mirroring actual societal behaviors.

Addressing these limitations, future extensions of this work present exciting opportunities for enriching our understanding of political polarization. Incorporating a multidimensional opinion vector could offer a more accurate representation of the complex landscape of individual ideologies, capturing the intricacies of political beliefs beyond a binary scale. Furthermore, expanding the model’s application to contexts beyond the United States would allow for a comparative analysis of polarization dynamics across different societal, cultural, and political ecosystems. Such an exploration could unveil the influence of varying frameworks on the processes and outcomes of polarization.

Transitioning the computational work to a programming language like C++ could dramatically reduce runtime, enabling a broader and more intricate analysis. This technical enhancement would facilitate a deeper dive into sensitivity and robustness tests and accommodate the inclusion of more complex dynamics within feasible computational limits.

The introduction of a Markov definition to formalize the model’s interactions and transitions opens new avenues for a rigorous mathematical analysis. This formalization could shed light on the model’s behavior under varying conditions, offering insights into the stability or escalation of polarization. Lastly, integrating a geolocal distribution of agents and aligning their interactions and opinion shifts with actual election outcomes could significantly improve the model’s empirical relevance. This approach would not only provide a pathway for empirical validation but also explore the impact of spatial and community dynamics on polarization and electoral behaviors.

In conclusion, while the current limitations of our approach define the scope of our initial findings, they also illuminate the path for future research endeavors. Each proposed extension represents a stepping stone towards a more comprehensive and nuanced understanding of political polarization, promising to bridge the gap between theoretical models and the complex realities of societal division.

4.3 Conclusion

In the realm of understanding political polarization through the lens of Agent-Based Modeling (ABM), this dissertation has ventured into a sophisticated simulation of societal divisions. Our findings illuminate the significant impact of exposure, elite influence, responsiveness, and economic inequality on the trajectory of polarization, offering a multifaceted view of its dynamics.

The ABM's capacity to encapsulate the complex interplay of individual behaviors and societal structures underlines the utility of such models in dissecting and potentially mitigating the forces driving polarization. Through sensitivity analysis, the direct relationship between exposure to like-minded groups and polarization was distinctly observed, underscoring the pernicious effect of echo chambers. Meanwhile, the mixed outcomes associated with responsiveness and elite influence highlight the intricate mechanisms at play, with independents emerging as pivotal actors capable of modulating these dynamics.

The robustness analyses further demonstrated the model's resilience and the critical role of initial conditions in shaping polarization outcomes. These analyses underscore the importance of considering the starting points of societal opinions and the potential for skewed distributions to accelerate polarization.

As we look forward, the limitations identified pave the way for meaningful extensions. Incorporating multidimensional opinion spaces, considering the role of external shocks, and enhancing the model with empirical validation are steps that can enrich the ABM's complexity and applicability. The potential for applying this model to diverse geopolitical contexts and integrating more sophisticated computational techniques promises to extend our understanding of polarization beyond the confines of this study.

Ultimately, this dissertation contributes to a deeper understanding of political polarization, offering insights into its drivers and potential mitigators. While acknowledging the limitations inherent in any modeling endeavor, the findings presented here provide a valuable foundation for future research aimed at unraveling the complex tapestry of societal divisions.

Appendix A

Sensitivity Analysis

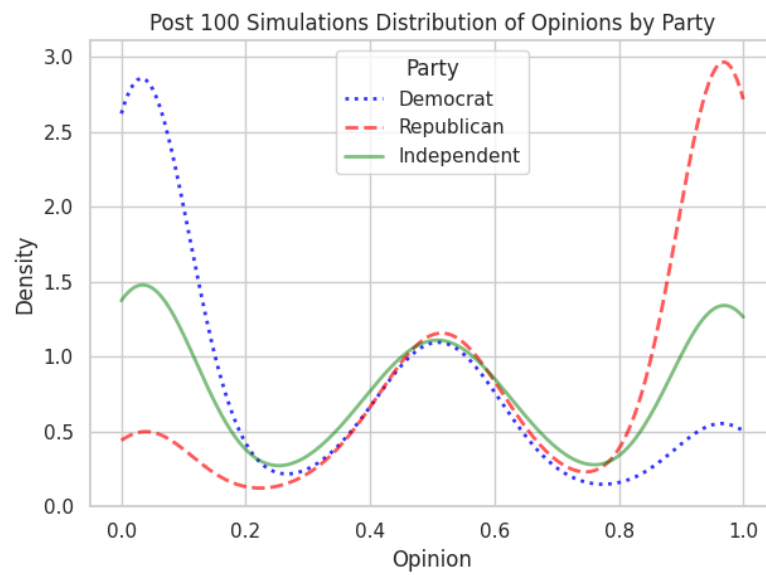


Figure A.1: Distribution of opinions by party post 100 simulations

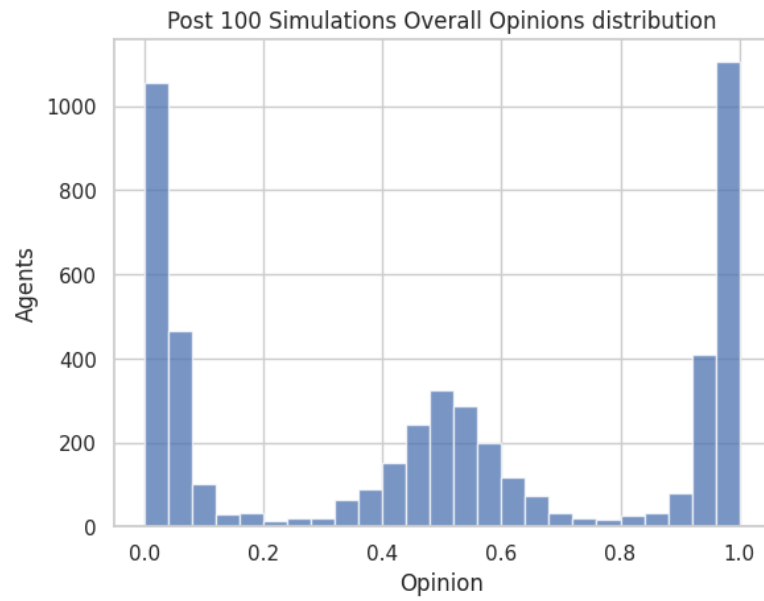


Figure A.2: Distribution of opinions aggregated post 100 simulations

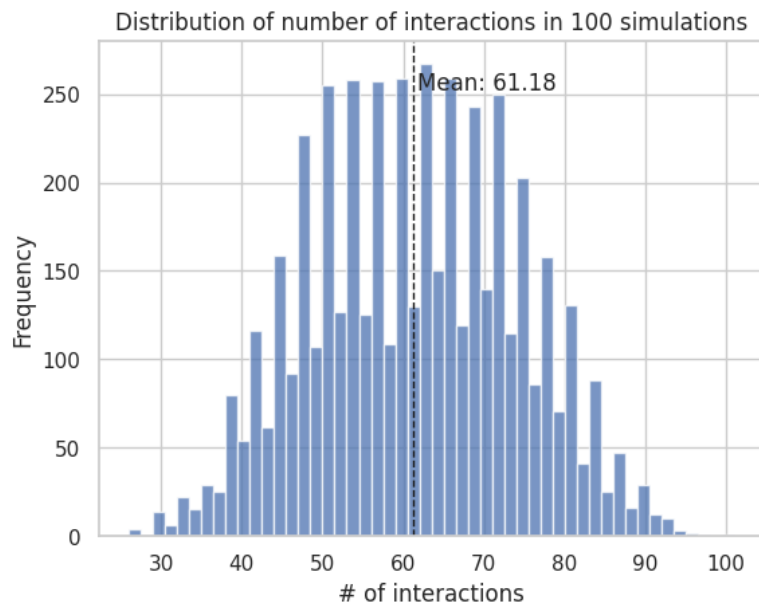


Figure A.3: Distribution of number of interactions in 100 simulations

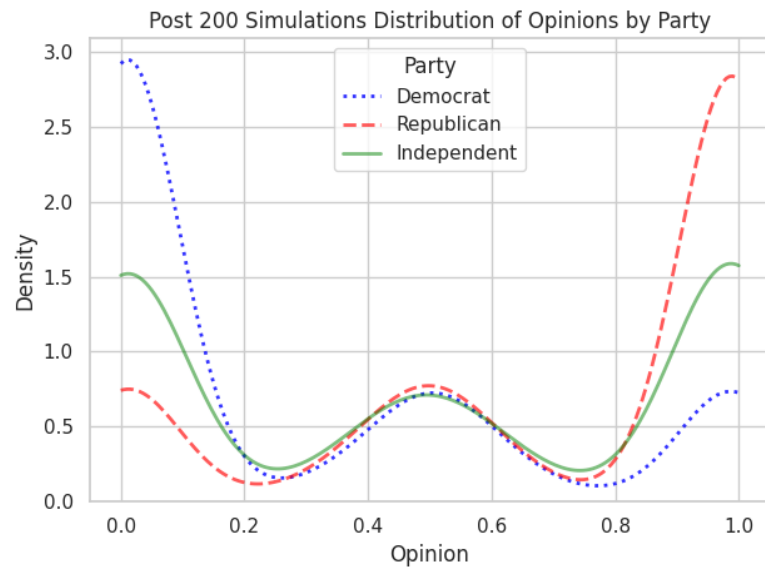


Figure A.4: Distribution of opinions by party post 300 simulations

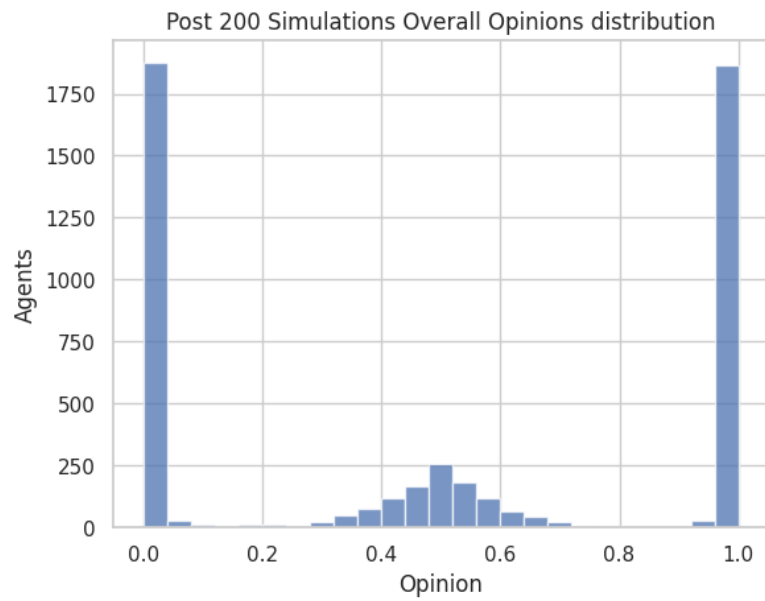


Figure A.5: Distribution of opinions aggregated post 100 simulations

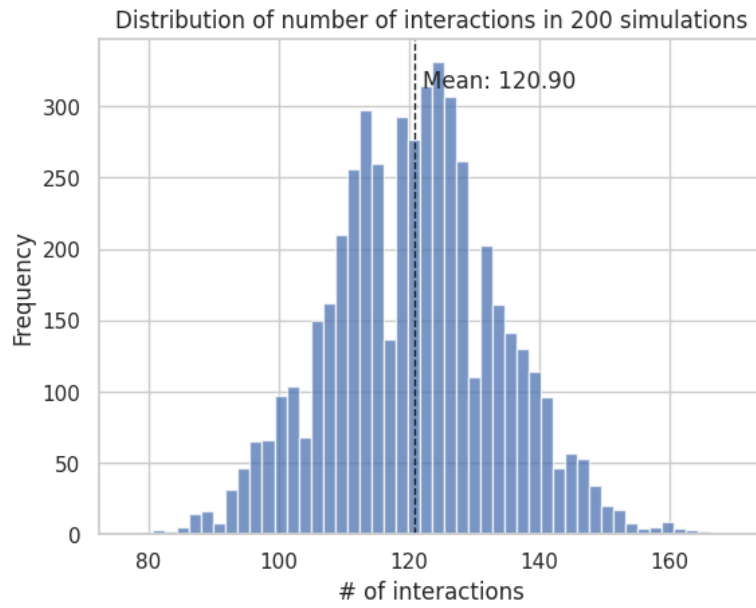


Figure A.6: Distribution of number of interactions in 200 simulations

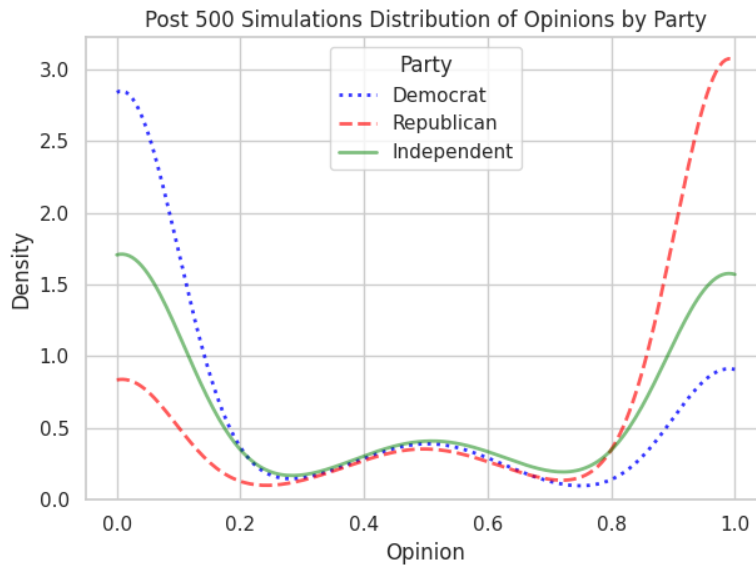


Figure A.7: Distribution of opinions by party post 100 simulations

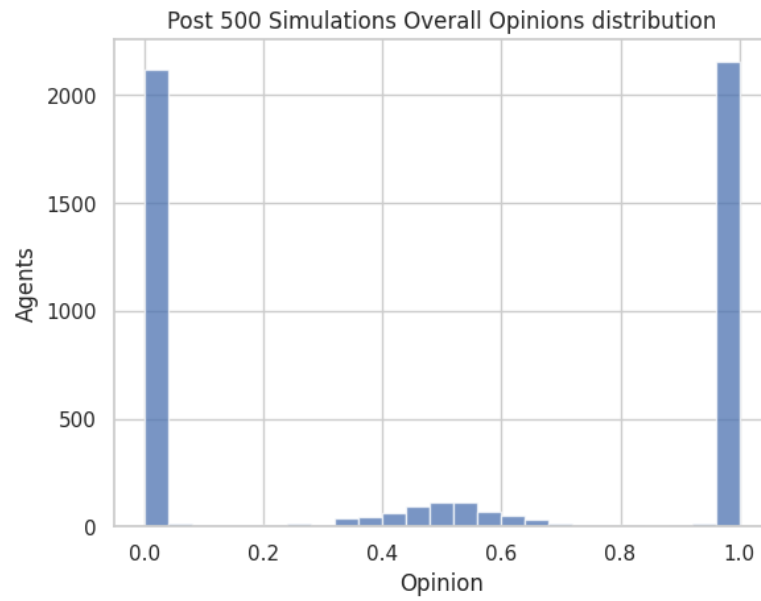


Figure A.8: Distribution of opinions aggregated post 100 simulations

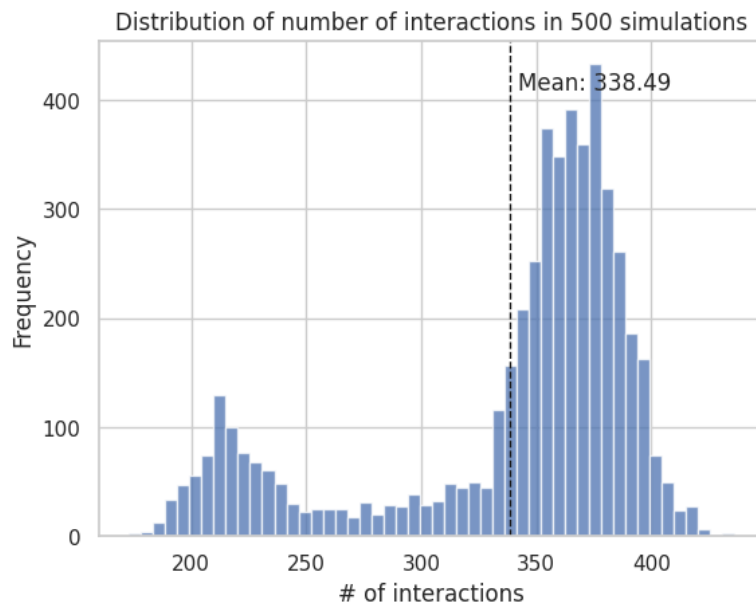


Figure A.9: Distribution of number of interactions in 500 simulations

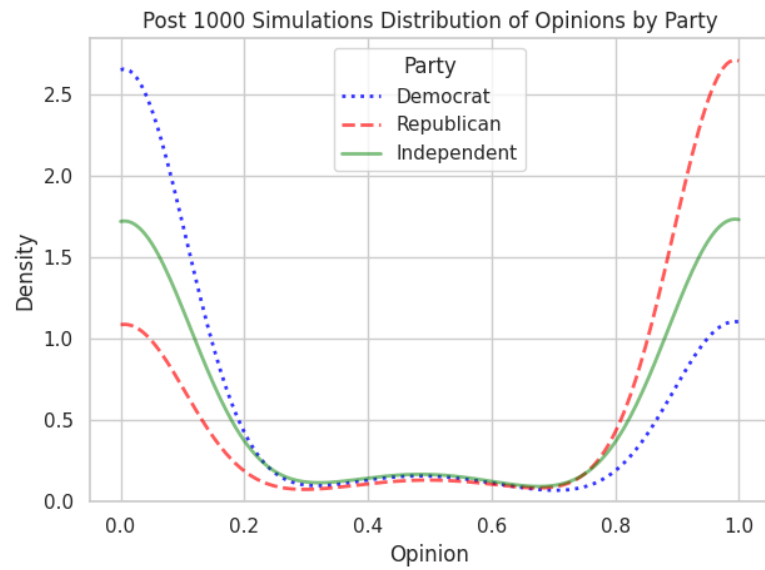


Figure A.10: Distribution of opinions by party post 1000 simulations

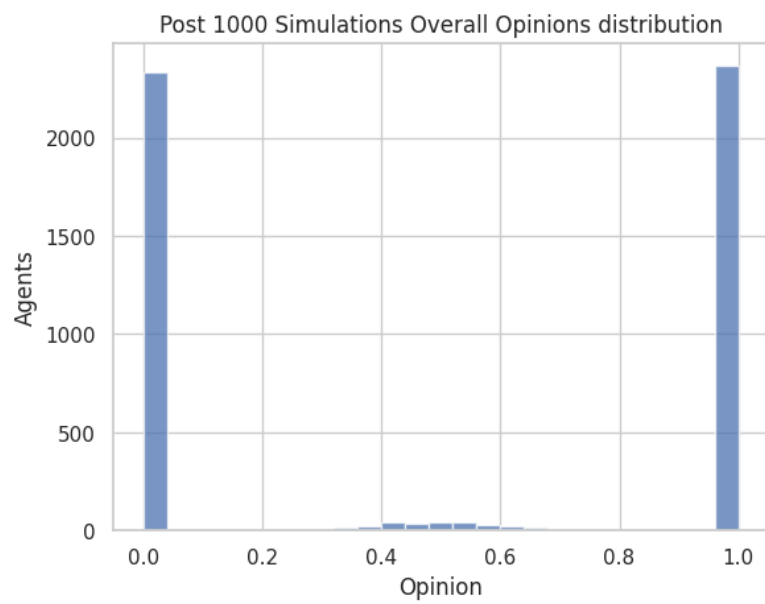


Figure A.11: Distribution of opinions aggregated post 1000 simulations

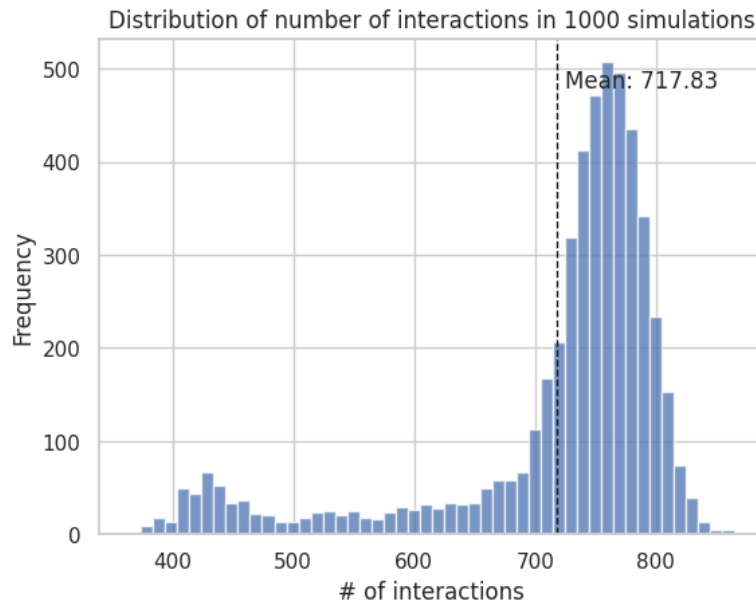


Figure A.12: Distribution of number of interactions in 1000 simulations

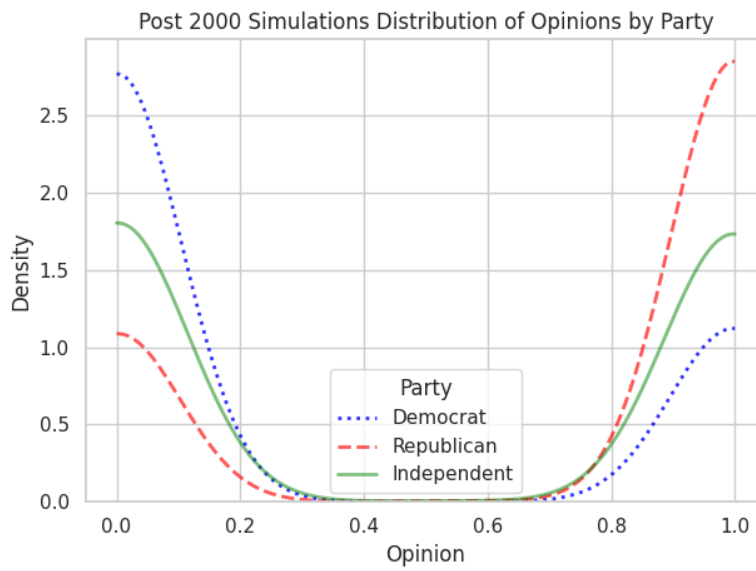


Figure A.13: Distribution of opinions by party post 2000 simulations

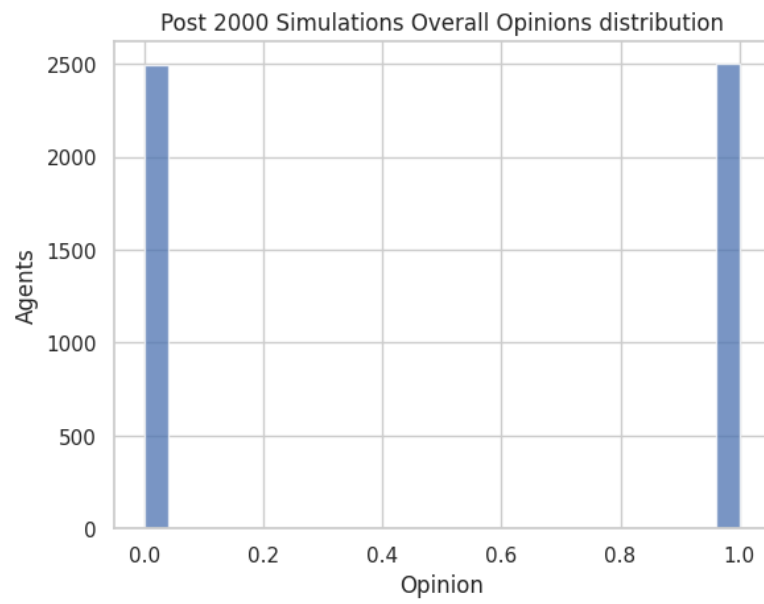


Figure A.14: Distribution of opinions aggregated post 2000 simulations

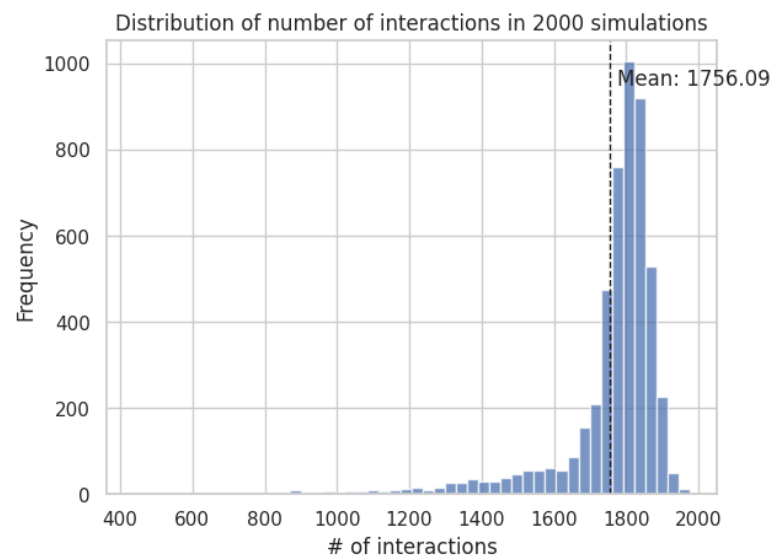


Figure A.15: Distribution of number of interactions in 2000 simulations

Appendix B

Robustness Analysis #1

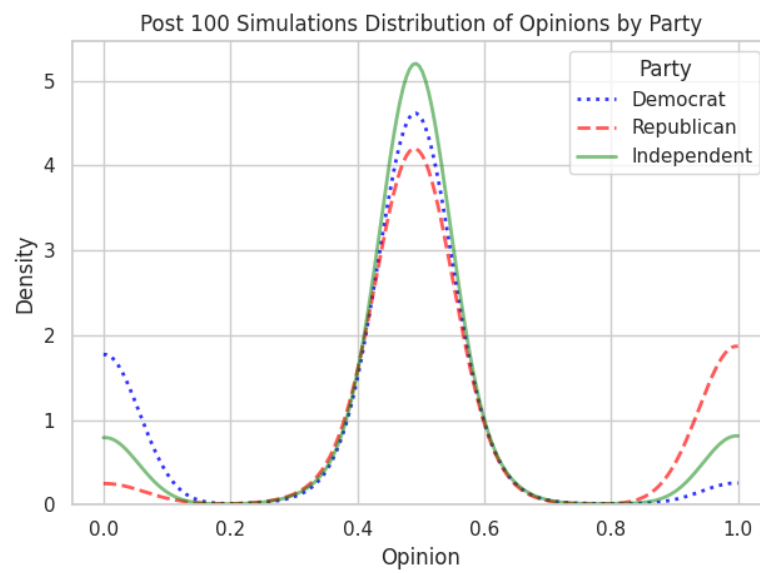


Figure B.1: Distribution of opinions by party post 100 simulations (*Less polarized*)

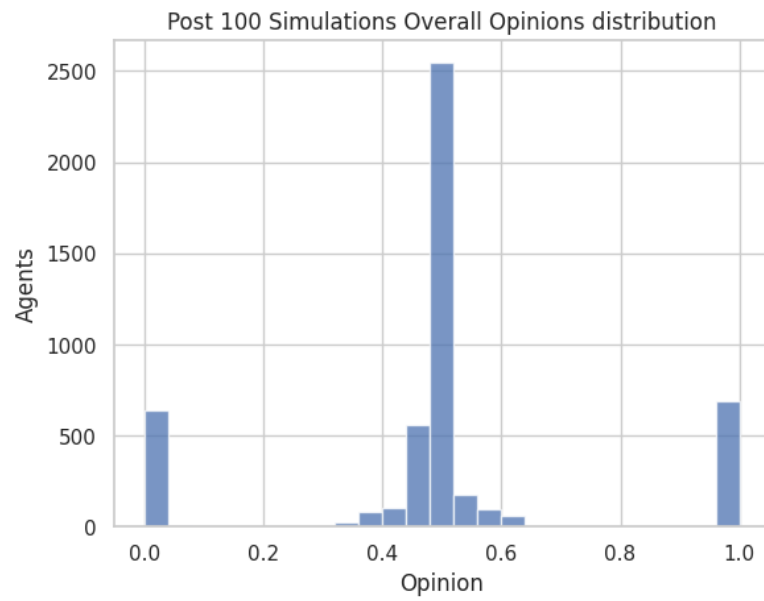


Figure B.2: Distribution of opinions aggregated post 100 simulations (*Less polarized*)

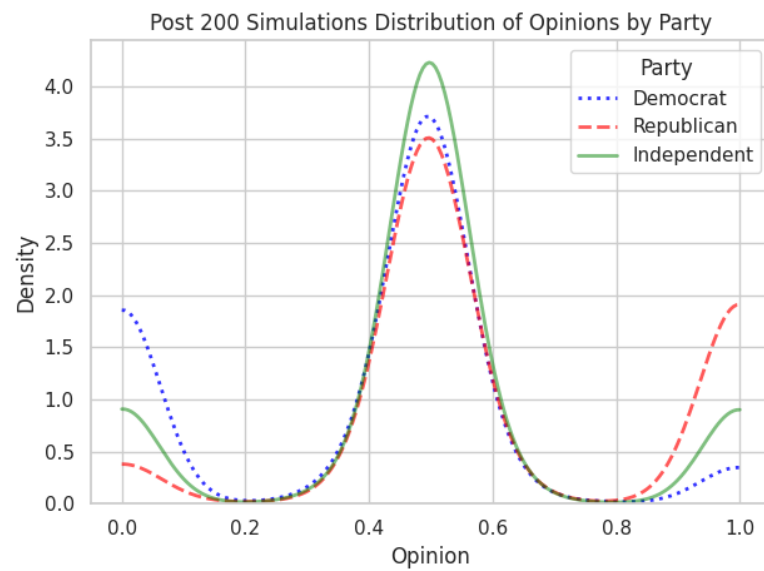


Figure B.3: Distribution of opinions by party post 200 simulations (*Less polarized*)

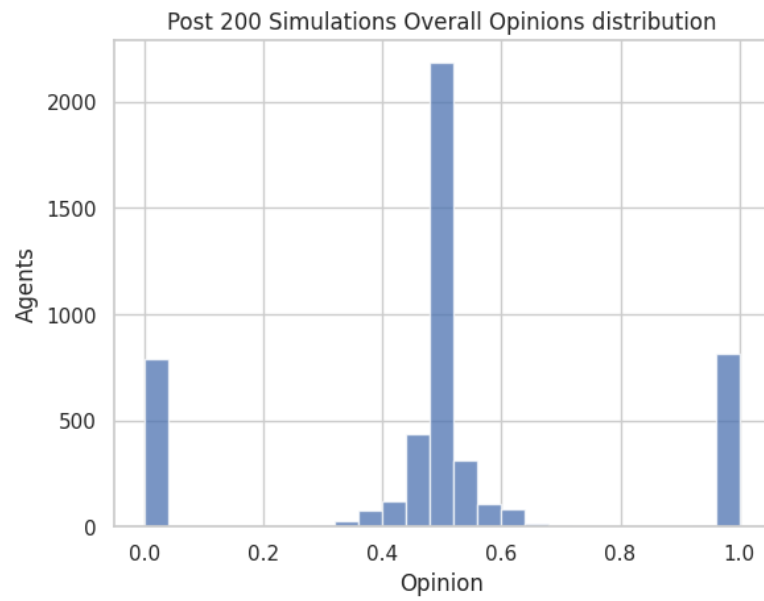


Figure B.4: Distribution of opinions aggregated post 200 simulations (*Less polarized*)

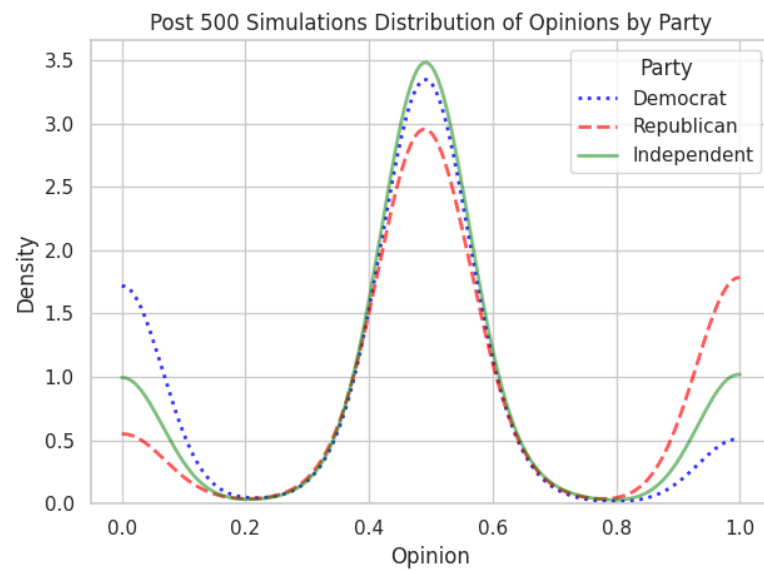


Figure B.5: Distribution of opinions by party post 500 simulations (*Less polarized*)

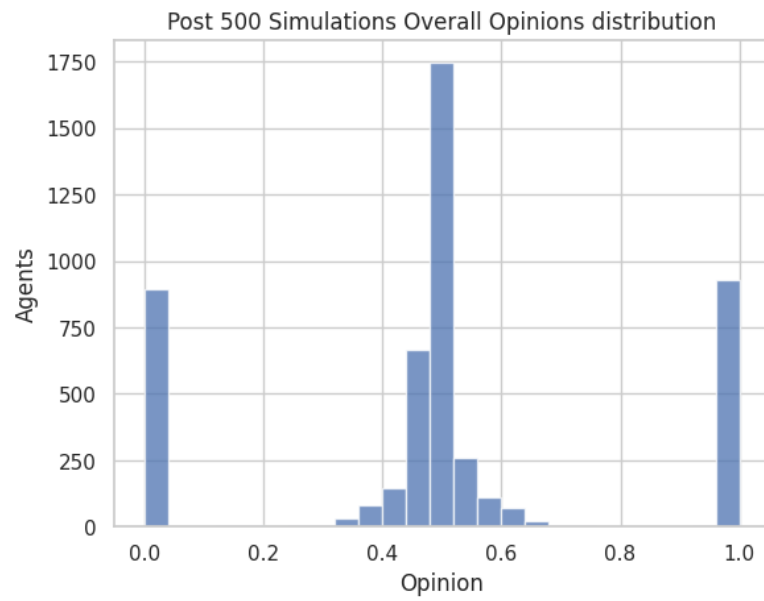


Figure B.6: Distribution of opinions aggregated post 500 simulations (*Less polarized*)

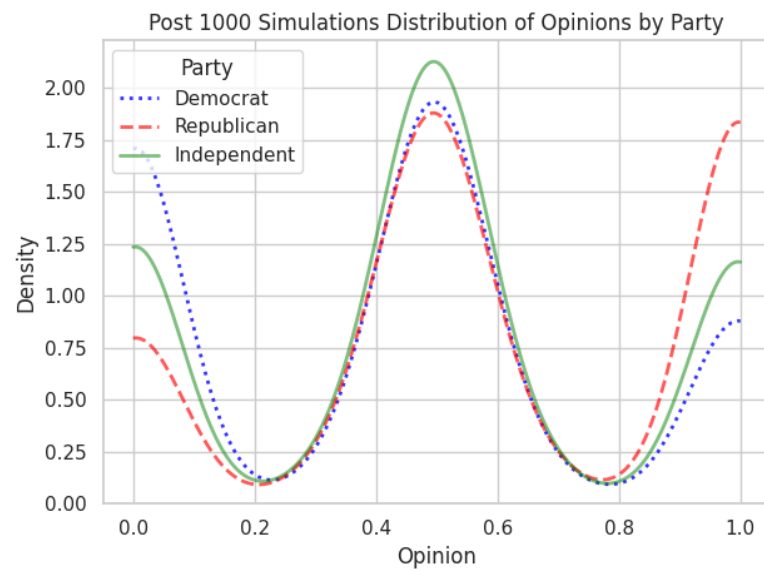


Figure B.7: Distribution of opinions by party post 1000 simulations (*Less polarized*)

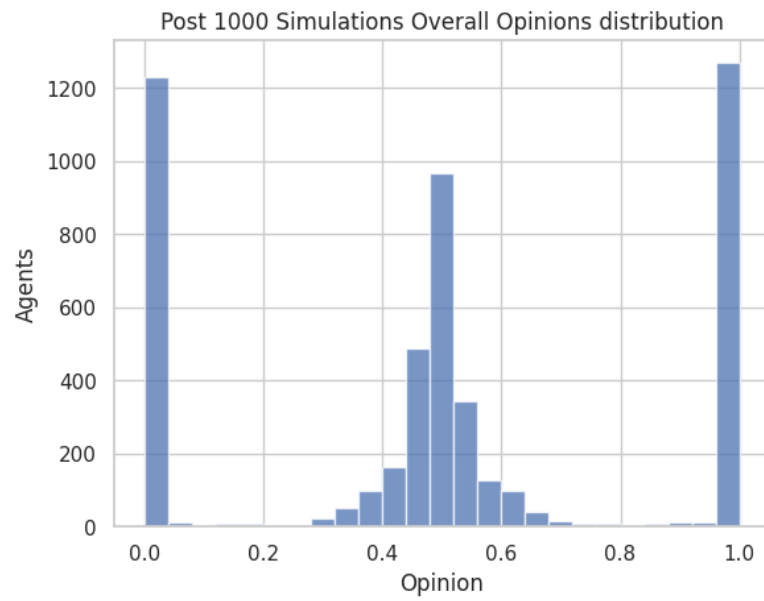


Figure B.8: Distribution of opinions aggregated post 1000 simulations (*Less polarized*)

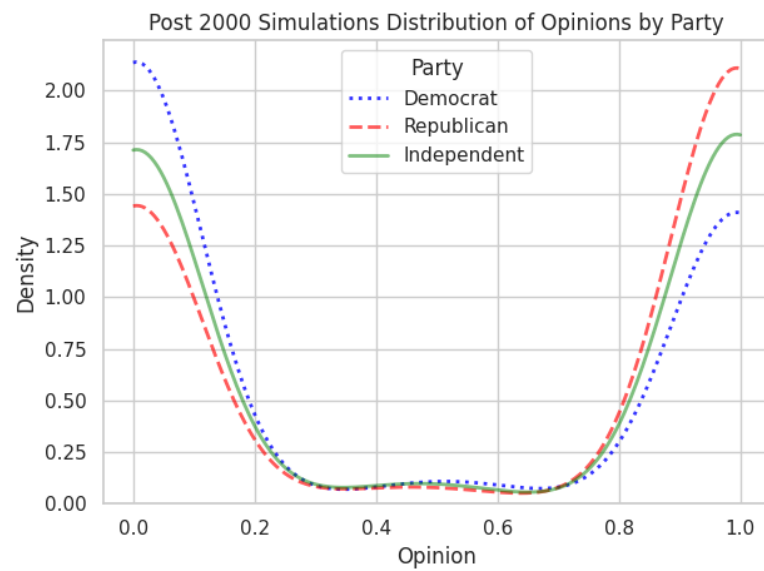


Figure B.9: Distribution of opinions by party post 2000 simulations (*Less polarized*)

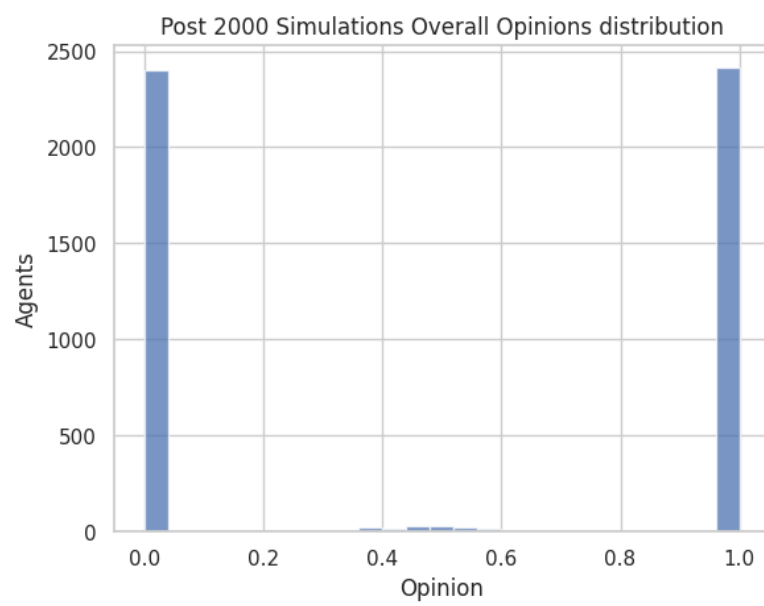


Figure B.10: Distribution of opinions aggregated post 2000 simulations (*Less polarized*)

Appendix C

Robustness Analysis #2

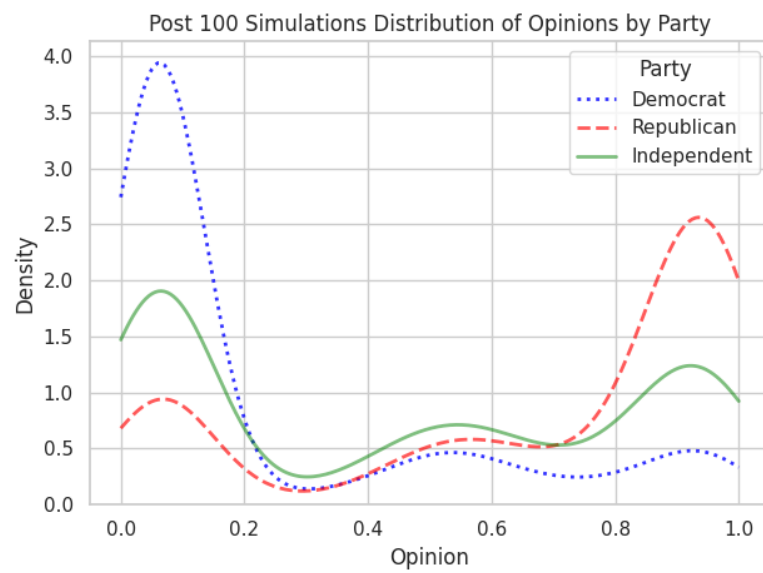


Figure C.1: Distribution of opinions by party post 100 simulations (*Skewed*)

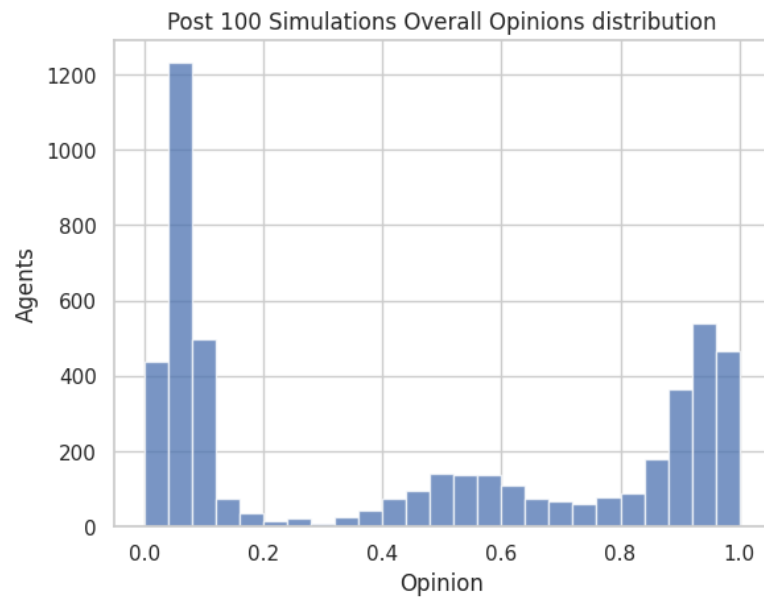


Figure C.2: Distribution of opinions aggregated post 100 simulations (*Skewed*)

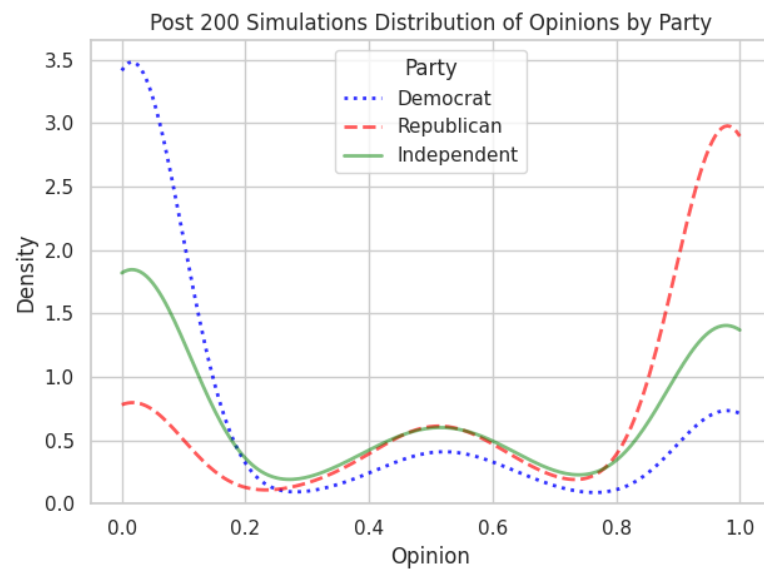


Figure C.3: Distribution of opinions by party post 200 simulations (*Skewed*)

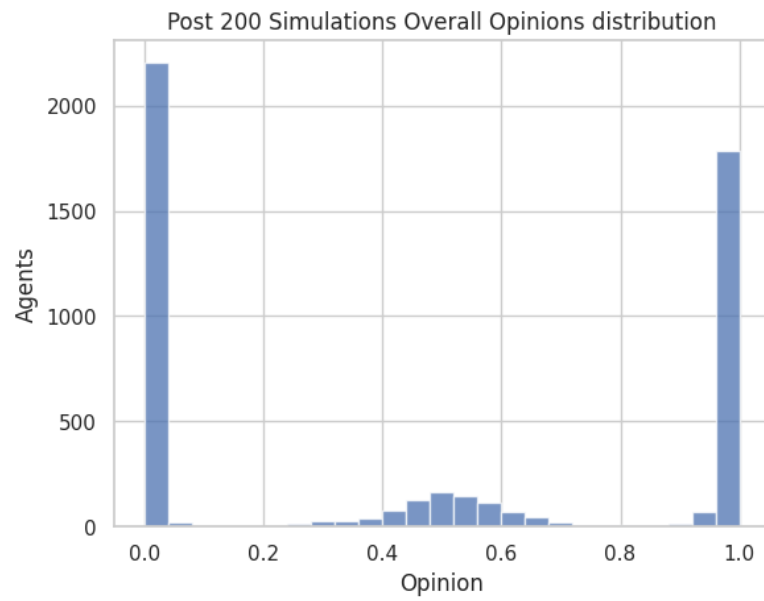


Figure C.4: Distribution of opinions aggregated post 200 simulations (*Skewed*)

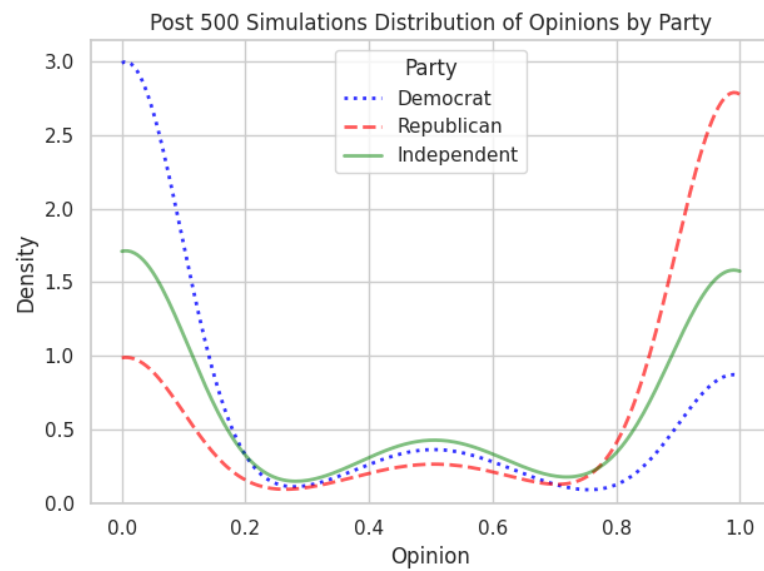


Figure C.5: Distribution of opinions by party post 500 simulations (*Skewed*)

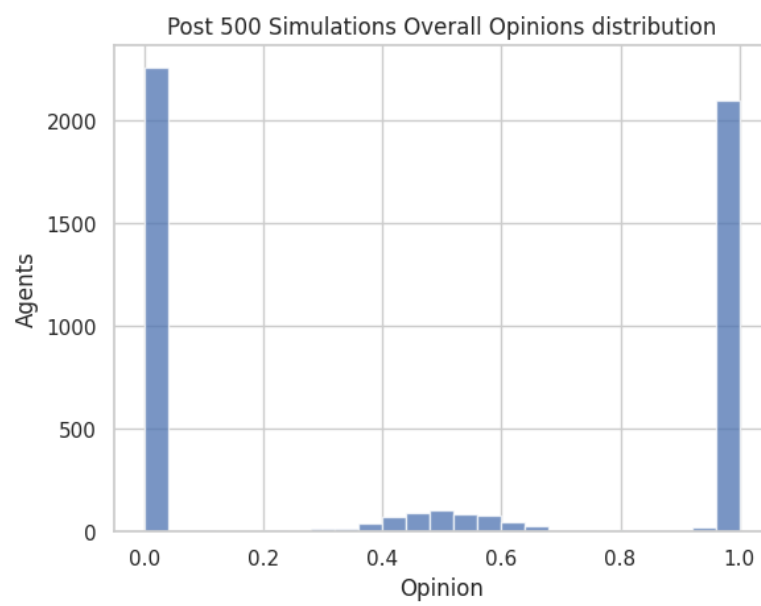


Figure C.6: Distribution of opinions aggregated post 500 simulations (*Skewed*)

Bibliography

- Political division in the united states, 2023. URL <https://www.statista.com/study/69077/political-division-in-the-united-states/>.
- ANES. The guide to public opinion and electoral behavior, 2020a. URL https://electionstudies.org/data-tools/anes-guide/anes-guide.html?chart=affective_polarization_parties.
- ANES. Wage inequality in the united states, 02 2024b. URL <https://www.statista.com/topics/3453/wage-inequality-in-the-united-states/#topicOverview>.
- Robert Axelrod, Joshua J. Daymude, and Stephanie Forrest. Preventing extreme polarization of political attitudes. *Proceedings of the National Academy of Sciences*, 118, 12 2021. doi: 10.1073/pnas.2102139118.
- Robert Axelrod, Stephanie Forrest, and Joshua J Daymude. Reply to de marchi: Modeling polarization of political attitudes. *Proceedings of the National Academy of Sciences of the United States of America*, 119, 04 2022. doi: 10.1073/pnas.2202863119.
- Robert M. Axelrod. *The complexity of cooperation : agent-based models of competition and collaboration*. Princeton University Press, 09 1997.
- Kevin K. Banda and John Cluverius. Elite polarization, party extremity, and affective polarization. *Electoral Studies*, 56:90–101, 12 2018. doi: 10.1016/j.electstud.2018.09.009.
- R. Benson, B. Duffy, R. Hesketh, and K. Hewlett. Attitudes to inequalities. <https://ifs.org.uk/inequality/wp-content/uploads/2021/09/Attitudes-to-inequalities.pdf>, 2021. Accessed: 2024-04-04.
- Dan Braha and Marcus A.M. de Aguiar. Voting contagion: Modeling and analysis of a century of u.s. presidential elections. *PLOS ONE*, 12:e0177970, 05 2017. doi: 10.1371/journal.pone.0177970.
- Hanna Bäck, Royce Carroll, Emma Renström, and Alexander Ryan. Elite communication and affective polarization among voters. *Electoral Studies*, 84:102639, 08

2023. doi: 10.1016/j.electstud.2023.102639. URL <https://researchportal.hkr.se/portalfiles/portal/63796949/1-s2.0-S0261379423000616-main.pdf>.
- Claudio Castellano, Santo Fortunato, and Vittorio Loreto. Statistical physics of social dynamics. *Reviews of Modern Physics*, 81:591–646, 05 2009. doi: 10.1103/revmodphys.81.591.
- Pew Research Center. As partisan divides over political values widen, other gaps remain more modest, 10 2017. URL https://www.pewresearch.org/politics/2017/10/05/1-partisan-divides-over-political-values-widen/1_3-19/. Accessed: 2024-03-26.
- I. Conjeaud, Philipp Lorenz-Spreen, and Argyris Kalogeratos. Degroot-based opinion formation under a global steering mechanism. *IEEE Transactions on Computational Social Systems*, pages 1–18, 2024. doi: 10.1109/TCSS.2023.3330293. doi: <https://doi.org/10.1109/tcss.2023.3330293>.
- N. Conte. Mapped: 2024 global elections by country. <https://www.visualcapitalist.com/2024-global-elections-by-country/>, 2024. Accessed: 20 Mar. 2024.
- Scott de Marchi. The complexity of polarization. *Proceedings of the National Academy of Sciences*, 119, 04 2022. doi: 10.1073/pnas.2115019119.
- M.H. DeGroot. Reaching a consensus. *Journal of the American Statistical Association*, 69(345):118–121, 1974. doi: 10.2307/2285509. doi: <https://doi.org/10.2307/2285509>.
- H T Diep, Miron Kaufman, and Sanda Kaufman. An agent-based statistical physics model for political polarization: A monte carlo study. *Entropy*, 25:981–981, 06 2023. doi: 10.3390/e25070981.
- James N. Druckman, Erik Peterson, and Rune Slothuus. How elite partisan polarization affects public opinion formation. *American Political Science Review*, 107:57–79, 01 2013. doi: 10.1017/s0003055412000500. URL <https://www.cambridge.org/core/journals/american-political-science-review/article/how-elite-partisan-polarization-affects-public-opinion-formation/6CB23BCCFBFB4EA3879D91232CEEA59>.
- Steven N Durlauf and H Peyton Young. *Social dynamics*. MIT Press, 2004.
- Adam M. Enders. Issues vs. affect: How do elite and mass polarization compare? *The Journal of Politics*, 05 2021. doi: 10.1086/715059.

- equalitytrust.org.uk. Trust, participation, attitudes and happiness | the equality trust. <https://equalitytrust.org.uk/trust-participation-attitudes-and-happiness>, n.d. Accessed: 2024-04-04.
- Yanfeng Gu and Zhongyuan Wang. Income inequality and global political polarization: The economic origin of political polarization in the world. *Journal of Chinese Political Science*, 27, 11 2021. doi: 10.1007/s11366-021-09772-1.
- Miron Kaufman, Sanda Kaufman, and Hung T. Diep. Statistical mechanics of political polarization. *Entropy*, 24:1262, 09 2022. doi: 10.3390/e24091262.
- Miron Kaufman, Sanda Kaufman, and Hung T Diep. Social depolarization: Blume–capel model. *Physics*, 6:138–147, 01 2024. doi: 10.3390/physics6010010.
- Ezra Klein. *Why We’re Polarized*. Avid Reader Press, 01 2020.
- Greg Lukianoff and Jonathan Haidt. *The Coddling of the American Mind: How Good Intentions and Bad Ideas Are Setting Up a Generation for Failure*. Penguin Books, 2019.
- Alexander V Mantzaris and George-Rafael Domenikos. Exploring the entropic nature of political polarization through its formulation as a isolated thermodynamic system. *Scientific Reports*, 13, 03 2023. doi: 10.1038/s41598.
- Gianluca Manzo. *Agent-based Models and Causal Inference*. John Wiley Sons, 01 2022.
- Nolan M Mccarty. *Polarization : what everyone needs to know*. Oxford University Press, 2019.
- Michael Mäs and Andreas Flache. Differentiation without distancing. explaining bi-polarization of opinions without negative influence. *PLoS ONE*, 8:e74516, 11 2013. doi: 10.1371/journal.pone.0074516.
- Martin A. Nowak. Five rules for the evolution of cooperation. *Science*, 314(5805):1560–1563, 2006. doi: 10.1126/science.1133755. URL <https://doi.org/10.1126/science.1133755>.
- ParthG60. Modelling Political Polarization. <https://github.com/ParthG60/Modelling-Political-Polarization/tree/main>, 2024. Accessed: 2024-04-05.

- Rasmus Skytte. Dimensions of elite partisan polarization: Disentangling the effects of incivility and issue polarization. *British Journal of Political Science*, 51:1–19, 06 2020. doi: 10.1017/s0007123419000760.
- Paul Smaldino. *Modeling Social Behavior*. Princeton University Press, 10 2023.
- Alexander J. Stewart, Nolan McCarty, and Joanna J. Bryson. Polarization under rising inequality and economic decline. *Science Advances*, 6:eabd4201, 12 2020. doi: 10.1126/sciadv.abd4201.
- Alexander J. Stewart, Joshua B. Plotkin, and Nolan McCarty. Inequality, identity, and partisanship: How redistribution can stem the tide of mass polarization. *Proceedings of the National Academy of Sciences*, 118, 12 2021. doi: 10.1073/pnas.2102140118.
- J.C. Turner and H. Tajfel. Social comparison and group interest in ingroup favouritism. *European Journal of Social Psychology*, 9(2):187–204, 1979. doi: 10.1002/ejsp.2420090207. URL <https://doi.org/10.1002/ejsp.2420090207>.
- Tim Urban. *What’s Our Problem? Wait But Why*, 02 2023.
- Vítor V. Vasconcelos, Simon A. Levin, and Flávio L. Pinheiro. Consensus and polarization in competing complex contagion processes. *Journal of The Royal Society Interface*, 16: 20190196, 06 2019. doi: 10.1098/rsif.2019.0196.
- Todd Washburn. Polarization and the political elite, 2022. URL <https://nationalaffairs.com/publications/detail/polarization-and-the-political-elite>.